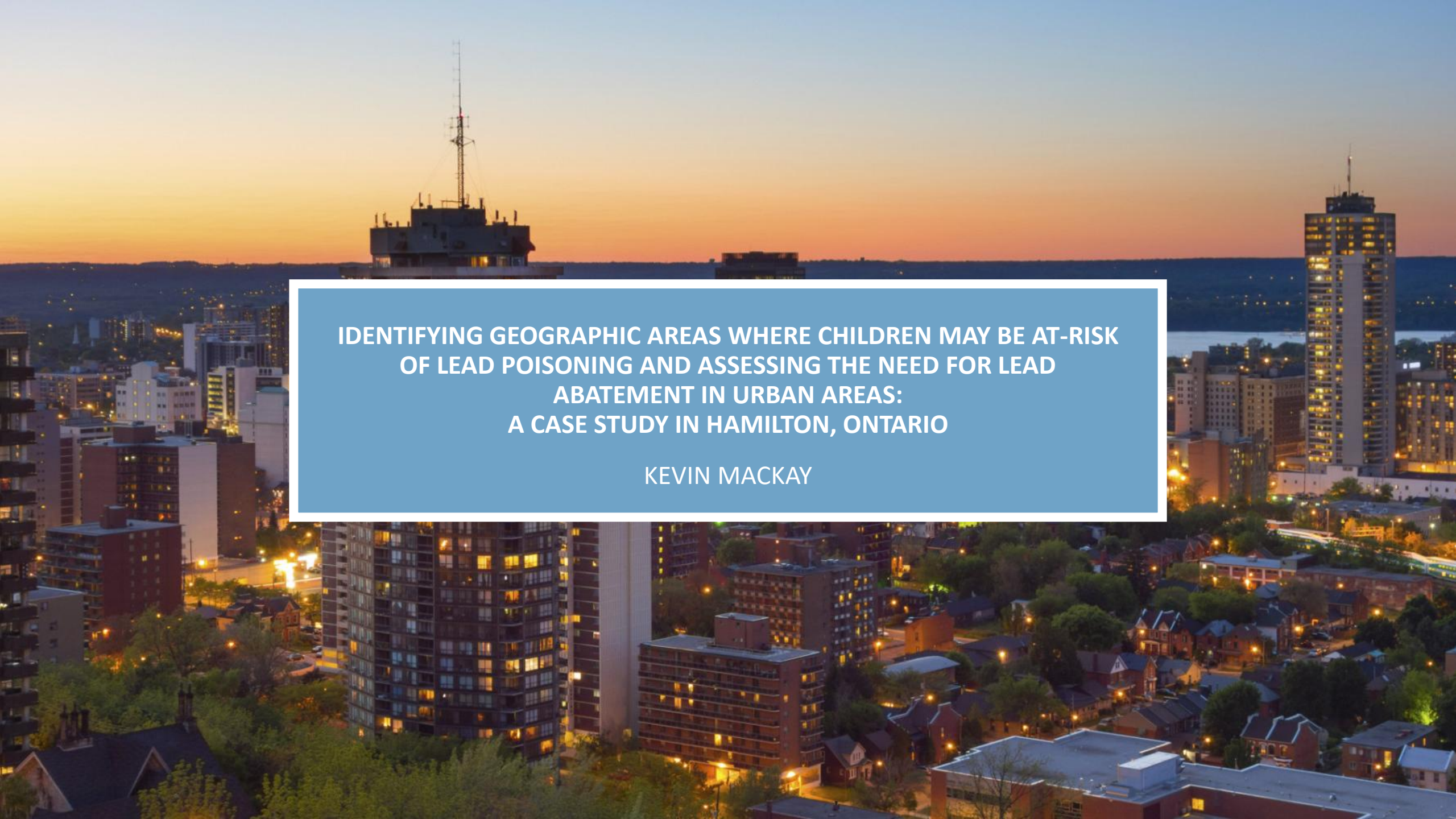


IDENTIFYING GEOGRAPHIC AREAS WHERE CHILDREN MAY BE AT-RISK OF LEAD POISONING AND ASSESSING THE NEED FOR LEAD ABATEMENT IN URBAN AREAS: A CASE STUDY IN HAMILTON, ONTARIO

KEVIN MACKAY

GOALS

1. Understand spatial distribution of SLLs
2. Understand BLL-IQ relationship
3. Understand necessity for lead abatement and ROI

CHAPTER 1

OBJECTIVE

Develop locally accurate model w/ LUR to interpolate SLLs

PROBLEM

Spurious and unintuitive relationships

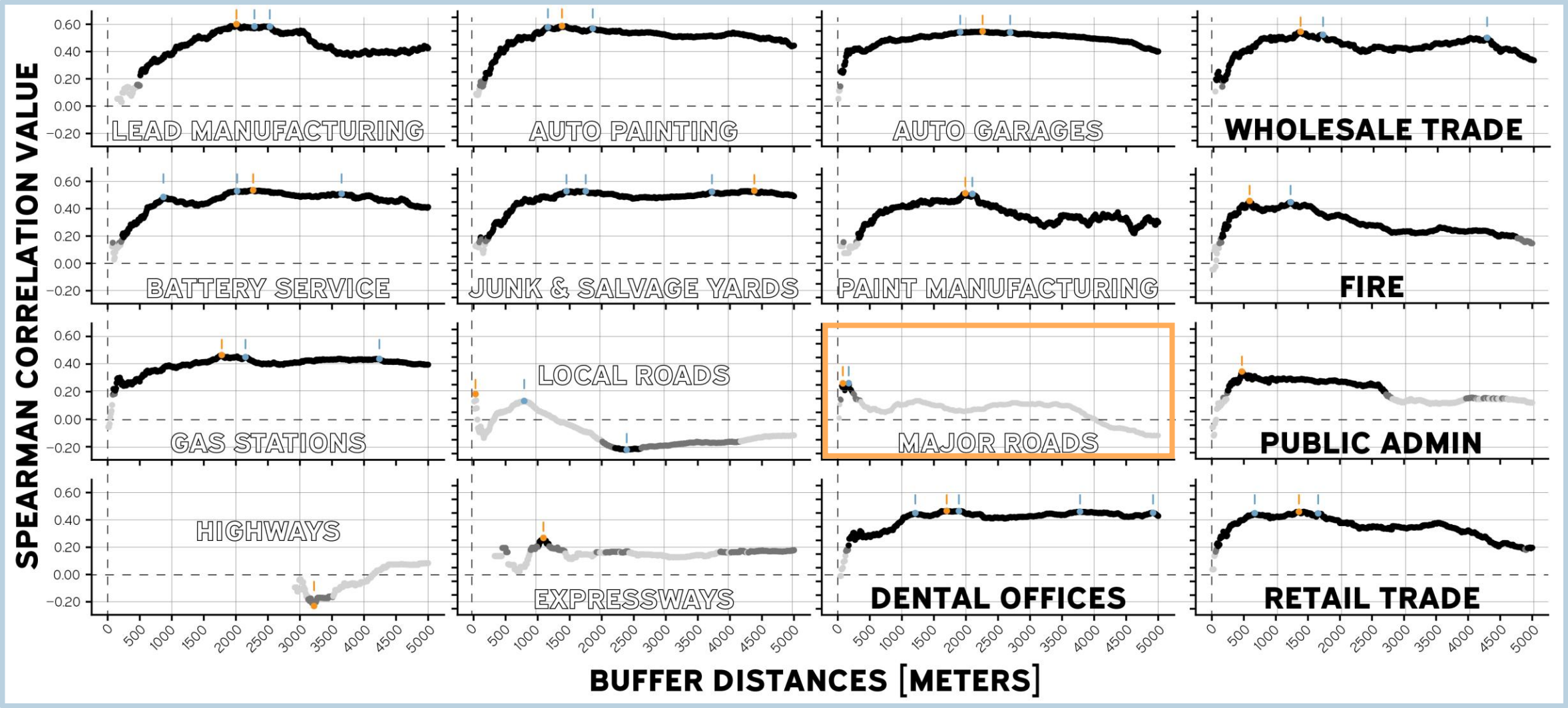
OUTCOME

Kriging outperforms Land Use Regression

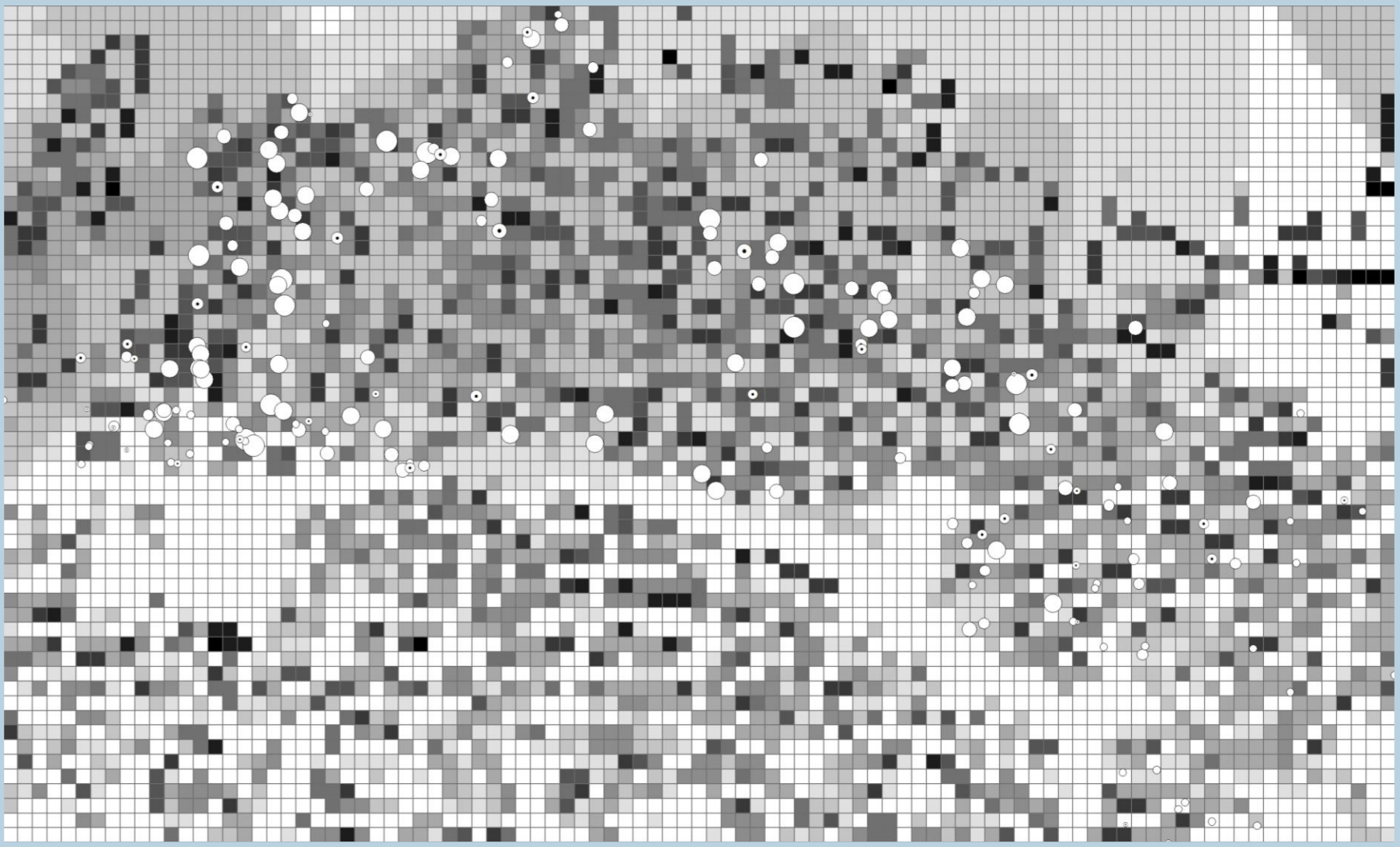
DISTANCE PROFILES

HISTORIC

CONTROL



OLV
ONLY LEAD-RELATED VARIABLES



SOIL LEAD LEVELS [mg/kg]

- 43.1 - 96.6
- 96.7 - 164.3
- 164.4 - 225.2
- 225.3 - 290.1
- 290.2 - 365.0
- 365.1 - 456.0
- 456.1 - 559.6
- 559.7 - 698.3
- 698.4 - 1007.2
- 1007.3 - 1842.

LUR MODEL SETS [mg/kg]

TRAINING SET	TEST SET
14.0 - 59.0	14.0 - 59.0
59.1 - 97.0	59.1 - 97.0
97.1 - 170.0	97.1 - 170.0
170.1 - 320.0	170.1 - 320.0
320.1 - 790.0	320.1 - 790.0

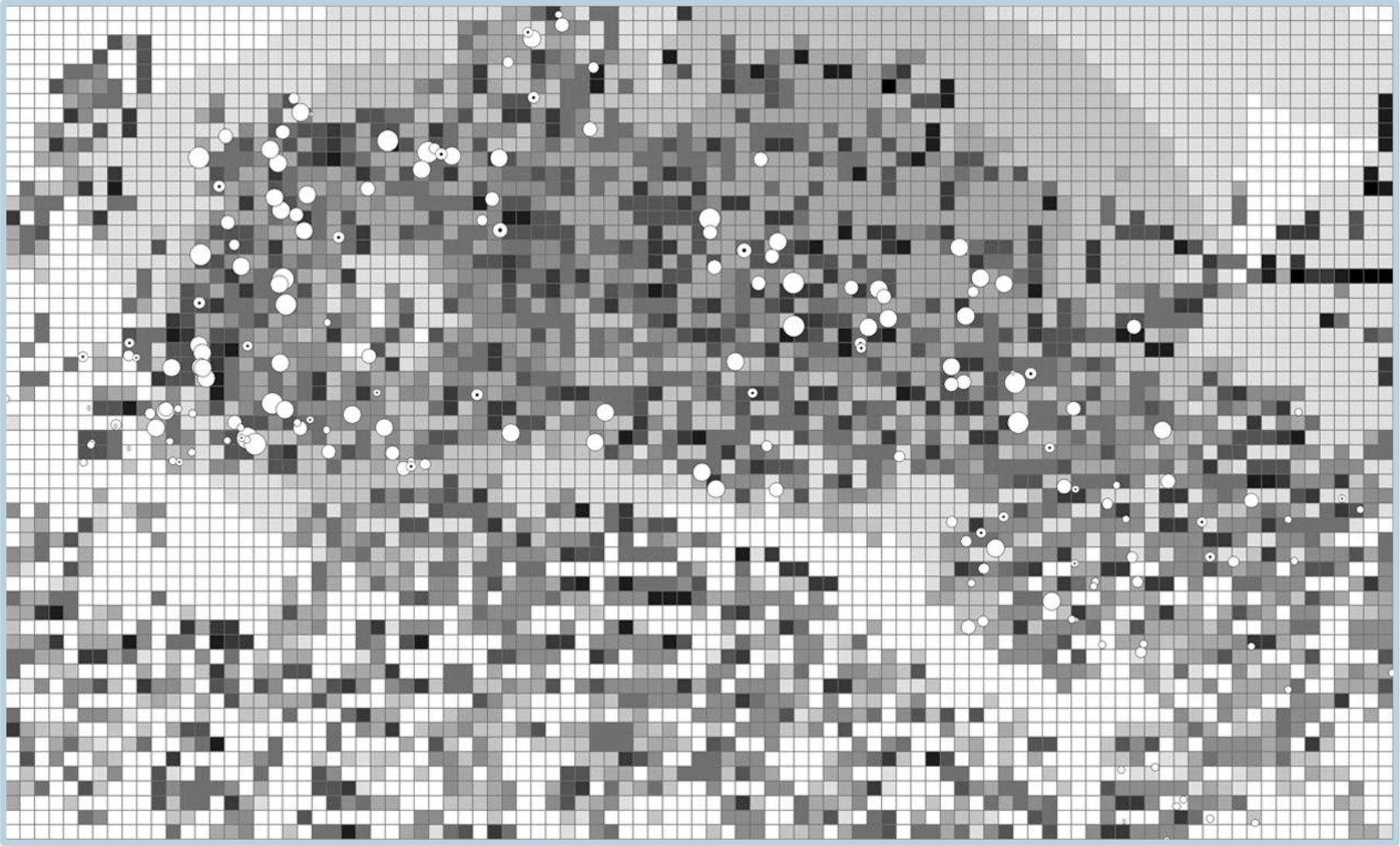
MODEL VARIABLES

LEAD MANUFACTURING
EXPRESSWAYS
LOCAL ROADS
AUTO PAINTERS

PERFORMANCE

ADJUSTED R-SQUARED: 0.24
RMSE: 209.0
AVERAGE ERROR: -141.8

AV
ALL VARIABLES



SOIL LEAD LEVELS [mg/kg]

- 43.1 - 96.6
- 96.7 - 164.3
- 164.4 - 225.2
- 225.3 - 290.1
- 290.2 - 365.0
- 365.1 - 456.0
- 456.1 - 559.6
- 559.7 - 698.3
- 698.4 - 1007.2
- 1007.3 - 1842.

LUR MODEL SETS [mg/kg]

TRAINING SET	TEST SET
14.0 - 59.0	14.0 - 59.0
59.1 - 97.0	59.1 - 97.0
97.1 - 170.0	97.1 - 170.0
170.1 - 320.0	170.1 - 320.0
320.1 - 790.0	320.1 - 790.0

MODEL VARIABLES

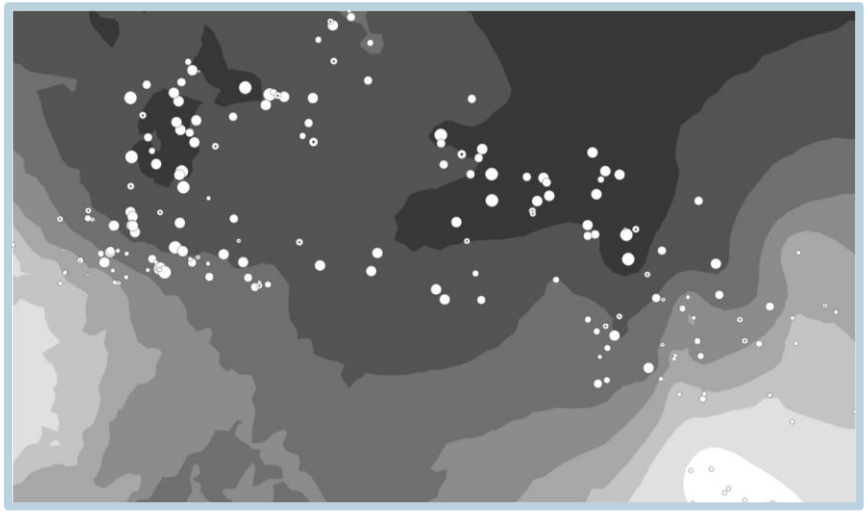
LEAD MANUFACTURING
WHOLESALE TRADE BLDGs
LOCAL ROADS
ADMINISTRATION BLDGs

PERFORMANCE

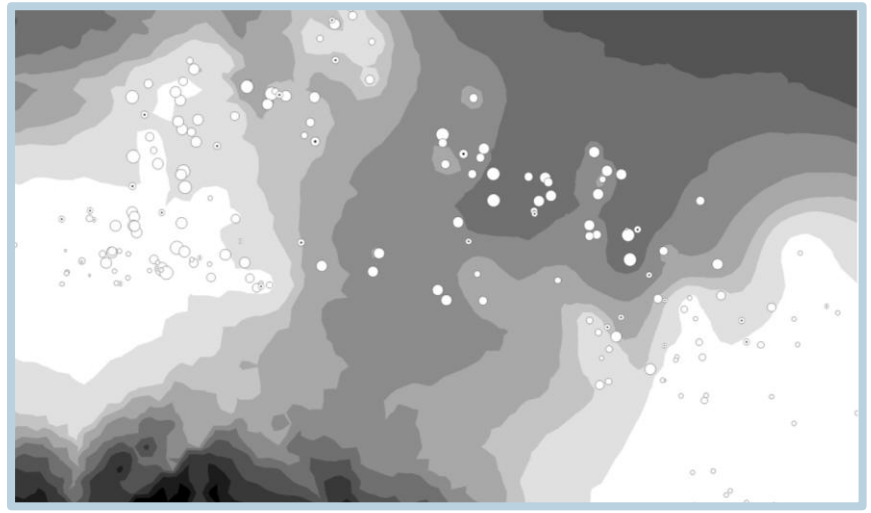
ADJUSTED R-SQUARED: 0.26
RMSE: 182.1
AVERAGE ERROR: -113.2

KRIGING

EBK | PREDICTIONS



EBK | STANDARD ERROR

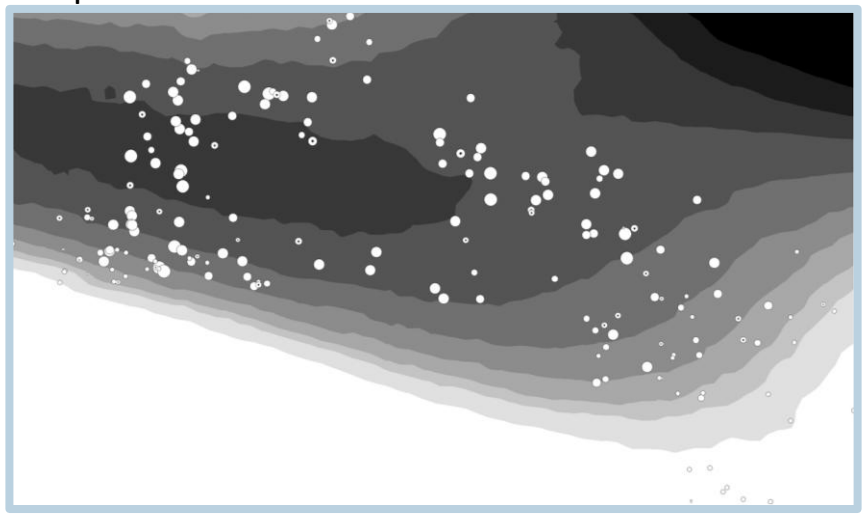


EBK | PERFORMANCE

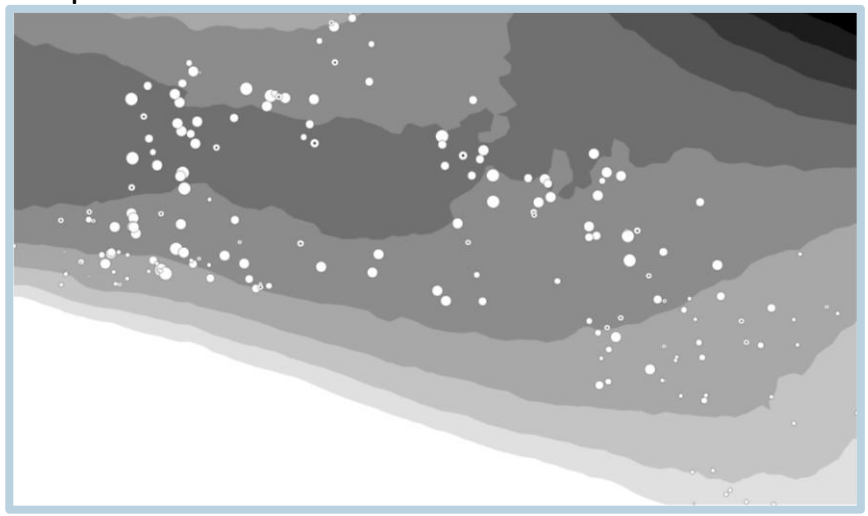
RMSE: 82.6

AVERAGE ERROR: -0.6

OK | PREDICTIONS



OK | STANDARD ERROR



OK | PERFORMANCE

RMSE: 76.7

AVERAGE ERROR: -7.1

SOIL LEAD LEVELS [mg/kg]

14 – 36.7	94.3 – 132.4
36.7 – 50.1	132.4 – 196.5
50.1 – 58.2	196.5 – 304.2
58.2 – 71.6	304.2 – 485.4
71.6 – 94.3	485.4 – 790

INTERPOLATION RESULTS

	OLV ONLY LEAD VARIABLES	AV ALL VARIABLES	EBK EMPIRICAL BAYESIAN KRIGING	OK ORDINARY KRIGING
MODEL VARIABLES	LEAD MANUFACTURING EXPRESSWAYS LOCAL ROADS AUTO PAINTERS	LEAD MANUFACTURING WHOLESALE TRADE BLDGs LOCAL ROADS ADMINISTRATION BLDGs	N/A	N/A
ADJUSTED R-SQUARED	0.24	0.26	N/A	N/A
RMSE	209.0	182.1	82.6	76.7
AVERAGE ERROR	-141.8	-113.2	-0.6	-7.1

CRITIQUES OF LUR

1. Resource-intensive
2. Risk of confounding factors
3. Risk of spurious relationships

WHY USE KRIGING?

1. Generalizable
2. Error estimation
3. Easy to implement
4. Resource management

CHAPTER 2

OBJECTIVE

Understand relationship between BLLs and IQ

METHOD

Using all available UBCs in two groups

OUTCOME

BLL-IQ relationship lower than previously thought

POOLED RESULTS WITH OUTLIERS REMOVED				
		GROUP	POOLED RESULT	I^2
	ALL UBCs	NON-LINEAR	-0.40	92.4%
		LINEAR	-0.20	47.7%
OUTLIERS REMOVED USING…	CONFIDENCE INTERVALS	NON-LINEAR	-0.73	93.8%
		LINEAR	-0.22	30.2%
	INFLUENCE ANALYSIS	NON-LINEAR	-0.64	70.7%
		LINEAR	-0.22	30.2%

SUBGROUP ANALYSIS

CHARACTERISTIC	GROUP	SUBGROUP	POOLED RESULT	TREND	I^2	SUBGROUP DIFFERENCE *P*-VALUE
UNADJUSTED VS. ADJUSTED	NON-LINEAR	UNADJUSTED	-0.80	↓	63.6%	0.14
		ADJUSTED	-0.52		73.3%	
	LINEAR	UNADJUSTED	-0.31	↓	41.8%	< 0.01
		ADJUSTED	-0.14		0.0%	
AGE OF CHILD	NON-LINEAR	INFANT	-0.22	↑	0.0%	< 0.01
		TODDLER	-0.59		57.5%	
		CHILD	-0.83		82.7%	
	LINEAR	INFANT	-0.03	↑	0.0%	< 0.01
		TODDLER	-0.19		12.3%	
		CHILD	-0.29		50.2%	

SUBGROUP ANALYSIS

CHARACTERISTIC	GROUP	SUBGROUP	POOLED RESULT	TREND	I^2	SUBGROUP DIFFERENCE *P*-VALUE
PRE-2009 VS. 2009-2019	NON-LINEAR	PRE-2009	-1.00	↓	77.1%	< 0.0001
		2009-2019	-0.29		45.0%	
	LINEAR	PRE-2009	-0.29	↓	39.1%	< 0.01
		2009-2019	-0.15		12.0%	
G7 COUNTRIES VS. OTHERS	NON-LINEAR	G7	-0.58	↑	50.9%	0.15
		OTHERS	-0.86		90.9%	
	LINEAR	G7	-0.21	↑	28.4%	0.25
		OTHERS	-0.50		64.8%	

PROJECTED LIFETIME SAVINGS PER COHORT				
SOURCE	IQ LOSS PER 1 µg/dL INCREASE IN BLLs [POINTS]	PROJECTED LIFETIME SAVINGS PER COHORT [MILLIONS of $CAD]	Discounted savings [Millions of $CAD]	
			8%	3%
HEALTH CANADA, 2013	1.00	$35,000	$1,500	$9,400
RESULTS OMITTING OUTLIERS	0.22	$7,700	$330	$2,100
ADJUSTED RESULT	0.14	$4,900	$210	$1,300
2009 to 2019 RESULT	0.15	$4,900	$210	$1,300
INFANTS	0.03	$169	$7	$45
TODDLERS	0.19	$4,400	$190	$1,200
CHILDREN	0.29	$1,700	$75	$469

CHAPTER 3

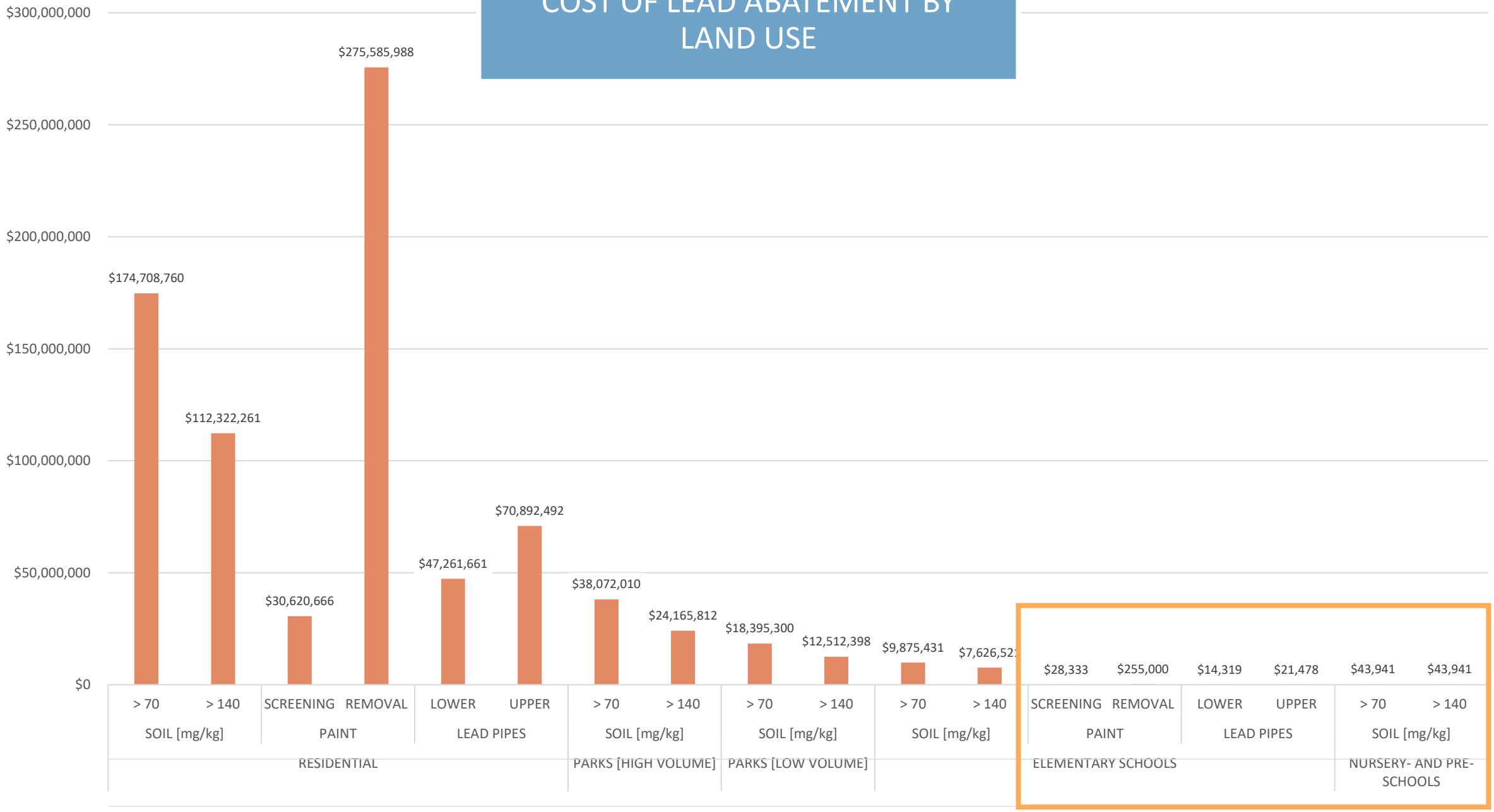
OBJECTIVE

Investigate cost and ROI for lead abatement in Hamilton

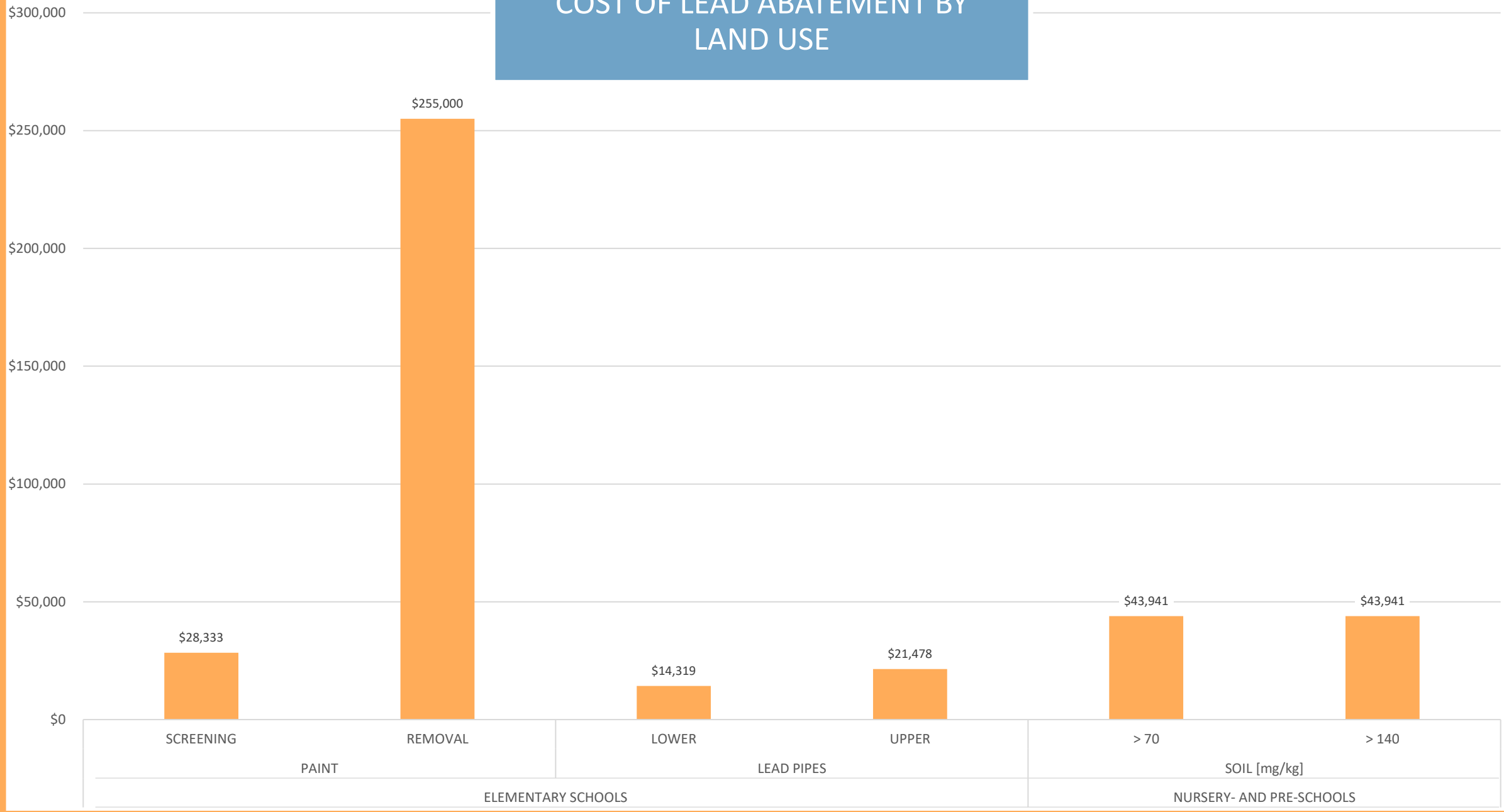
OUTCOME

Upfront capital cost high and ROI uncertain

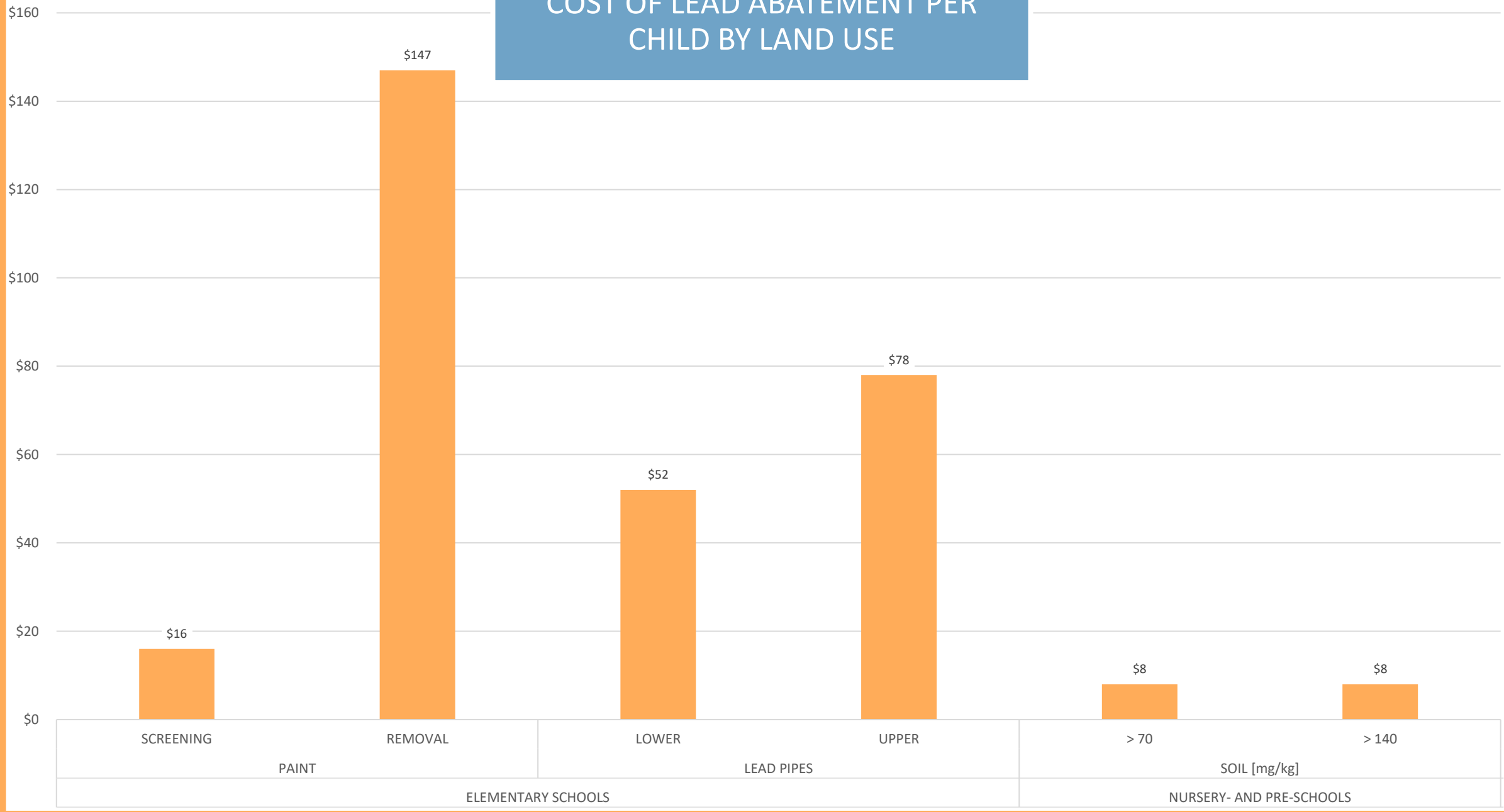
COST OF LEAD ABATEMENT BY LAND USE



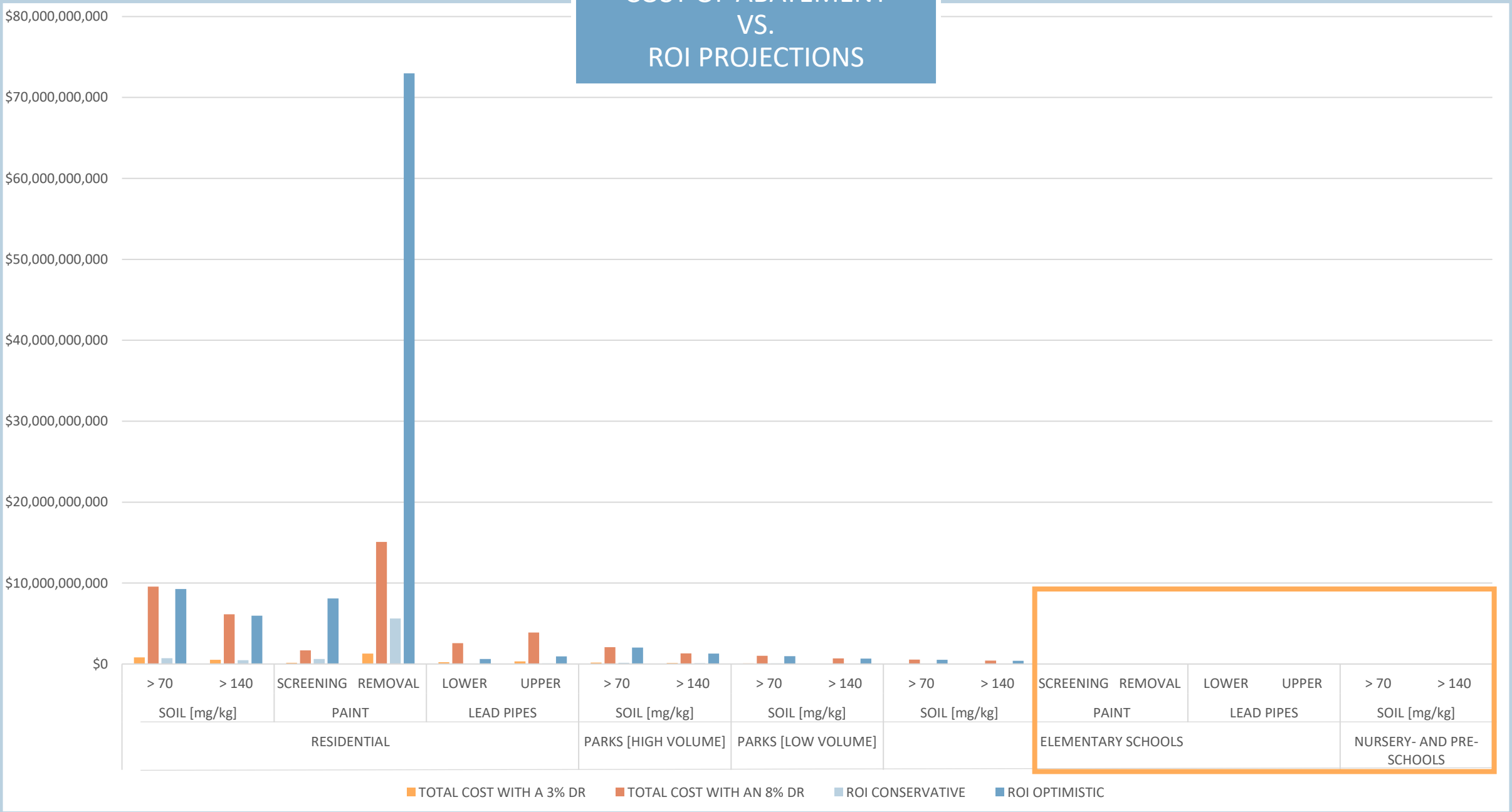
COST OF LEAD ABATEMENT BY LAND USE



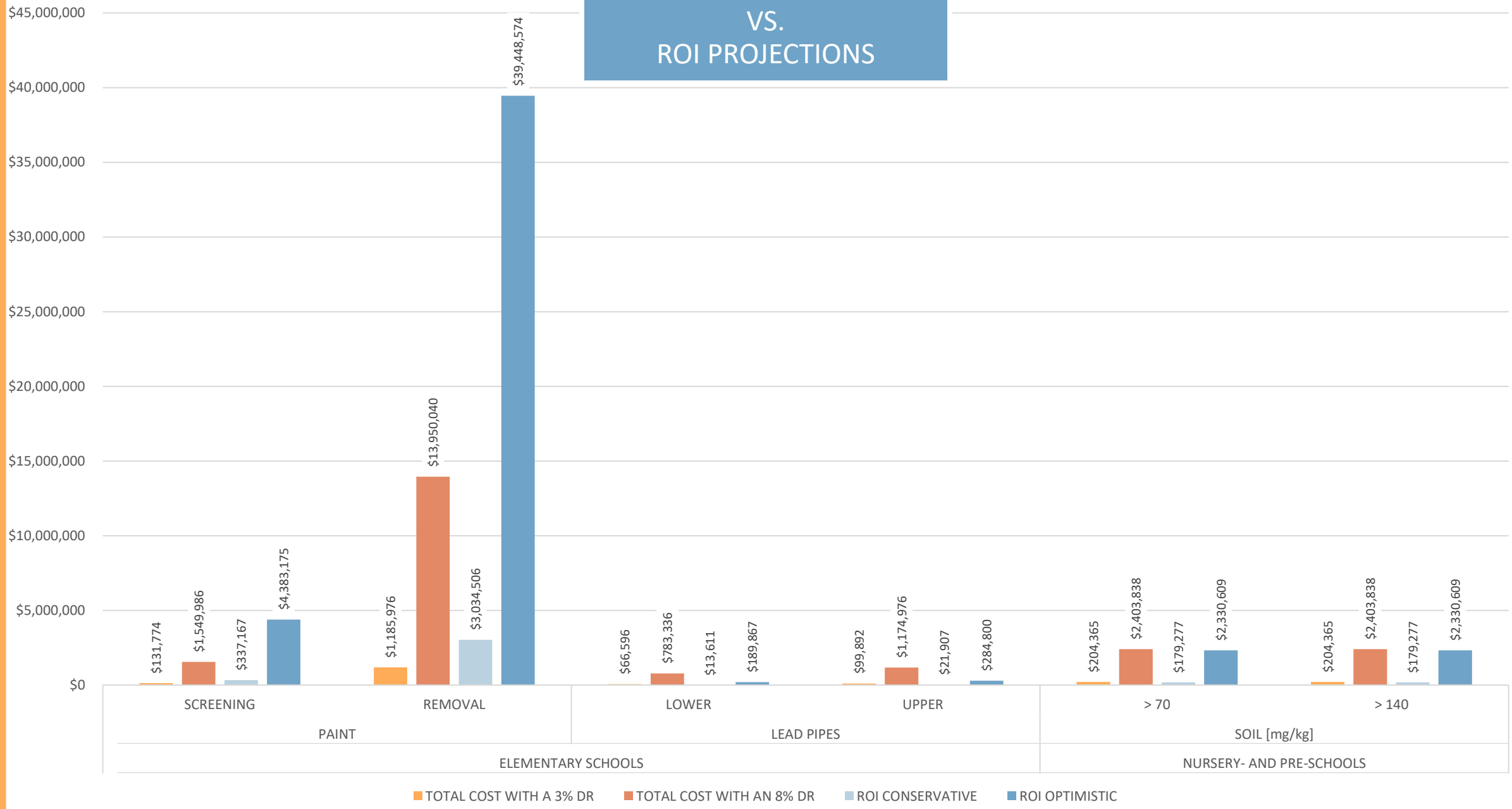
COST OF LEAD ABATEMENT PER CHILD BY LAND USE



COST OF ABATEMENT VS. ROI PROJECTIONS

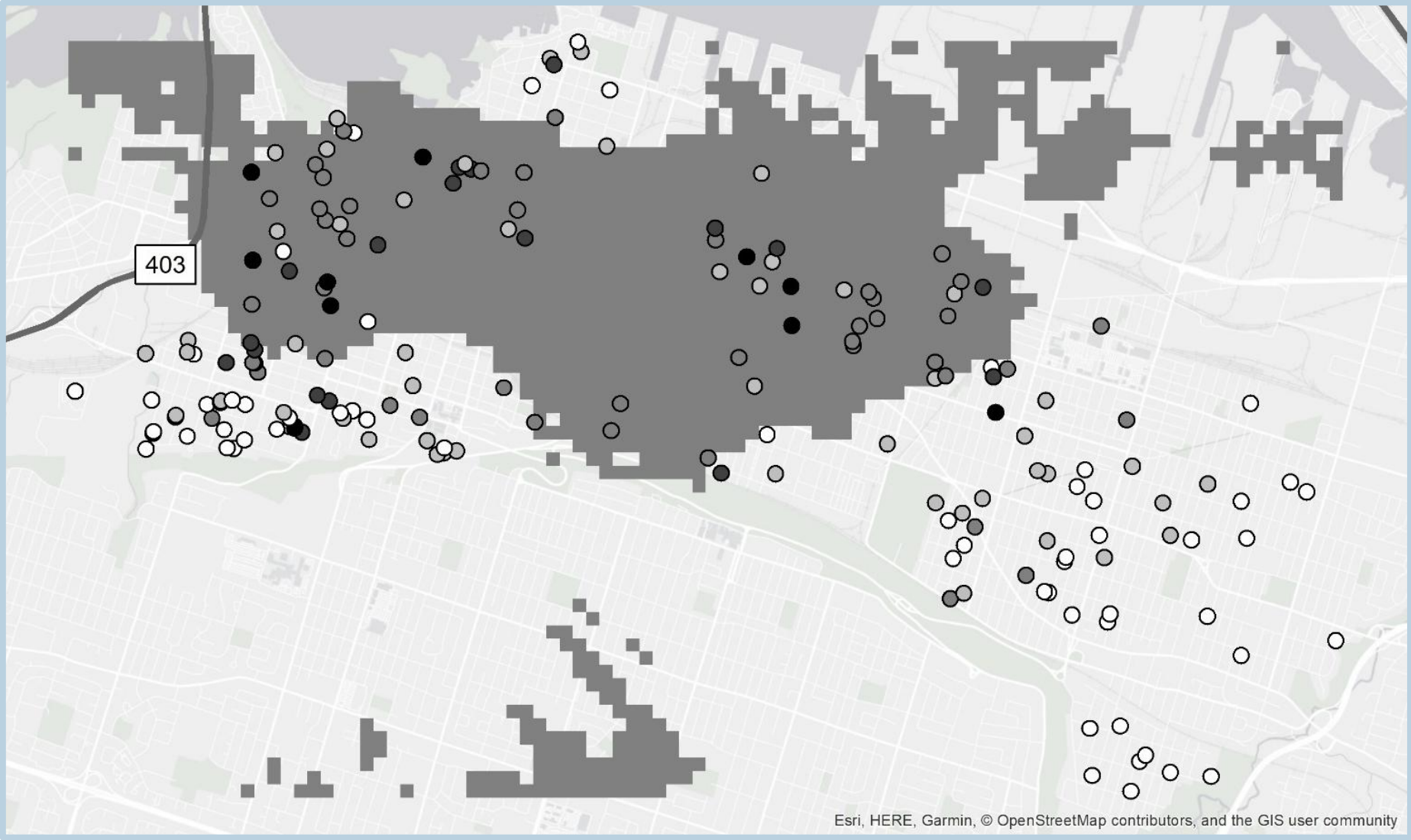


COST OF ABATEMENT VS. ROI PROJECTIONS



HAZARD QUOTIENT				
AGE GROUP	52 WEEKS OF EXPOSURE		40 WEEKS OF EXPOSURE	
	MAXIMUM	MEAN	MAXIMUM	MEAN
INFANT 0 - 0.5 years	0.396	0.195	0.304	0.150
TODDLER 0.6 - 4 years	0.786	0.388	0.605	0.299
CHILD 5 - 11 years	0.099	0.049	0.076	0.037
HQ < 0.5	VERY UNLIKELY TO EXCEED 5 µg/dL			
0.5 > HQ > 1.0	UNLIKELY TO EXCEED 5 µg/dL			
HQ > 1.0	MIGHT EXCEED 5 µg/dL			

TODDLERS WITH 52 WEEKS OF EXPOSURE



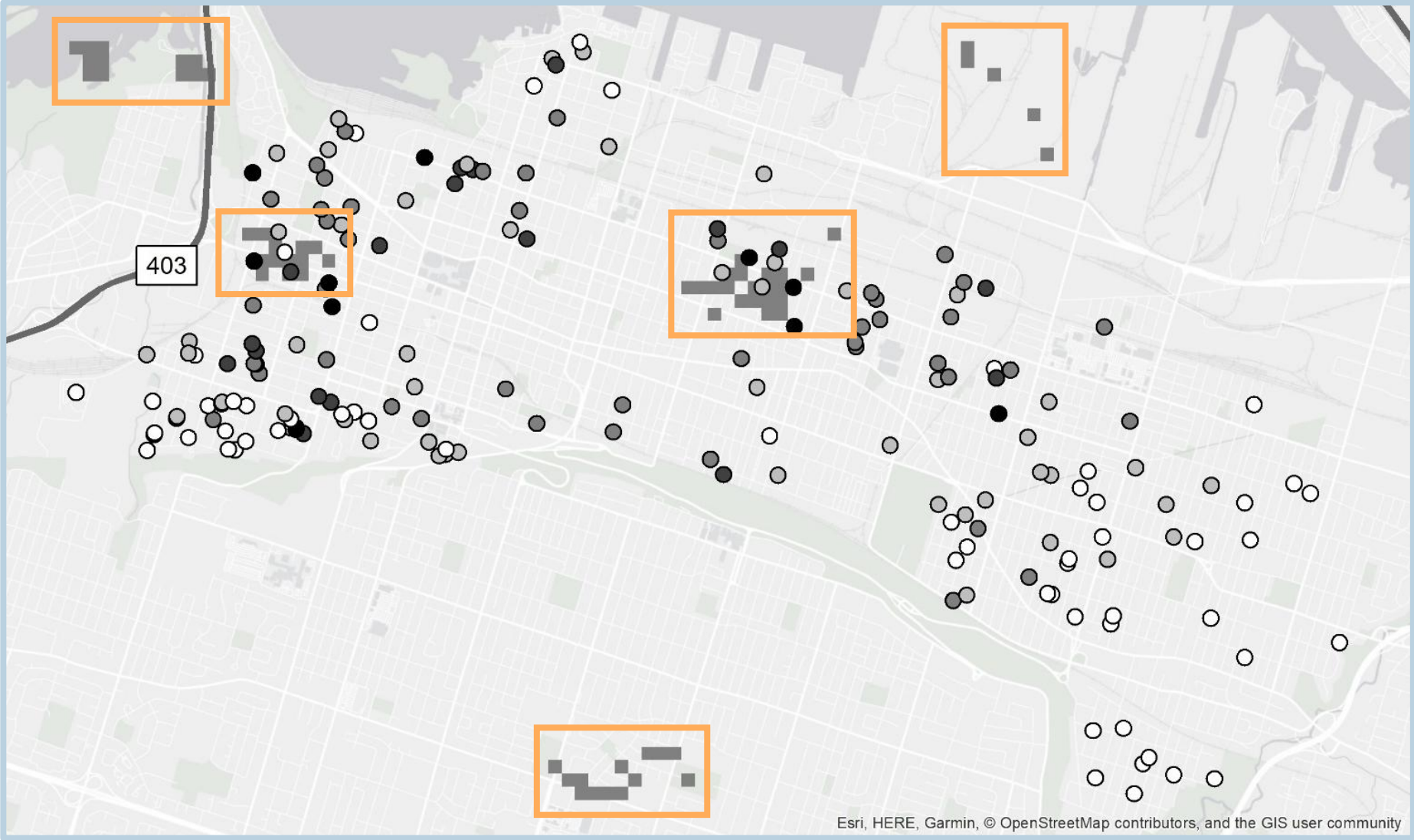
CHANCE OF A TODDLER EXCEEDING 5 µg/dL

- VERY UNLIKELY
- UNLIKELY
- MIGHT

SOIL LEAD LEVELS [mg/kg]

- ≤77.00
- ≤150.00
- ≤240.00
- ≤370.00
- ≤790.00

TODDLERS WITH 40 WEEKS OF EXPOSURE



CHANCE OF A TODDLER EXCEEDING 5 µg/dL

- VERY UNLIKELY
- UNLIKELY
- MIGHT

SOIL LEAD LEVELS [mg/kg]

- ≤77.00
- ≤150.00
- ≤240.00
- ≤370.00
- ≤790.00

DISTRIBUTION OF ROI SAVINGS				
SAVINGS	MUNICIPAL	PROVINCIAL	FEDERAL	NOTES
HEALTH CARE		X		
LIFETIME EARNINGS	X	X	X	Depends if person lives in city or moves away
TAX REVENUE	X	X	X	
CRIME	X			
SPECIAL EDUCATION		X		
VIOLENT CRIME		X	X	Longer sentences w/ higher security

CONCLUSIONS

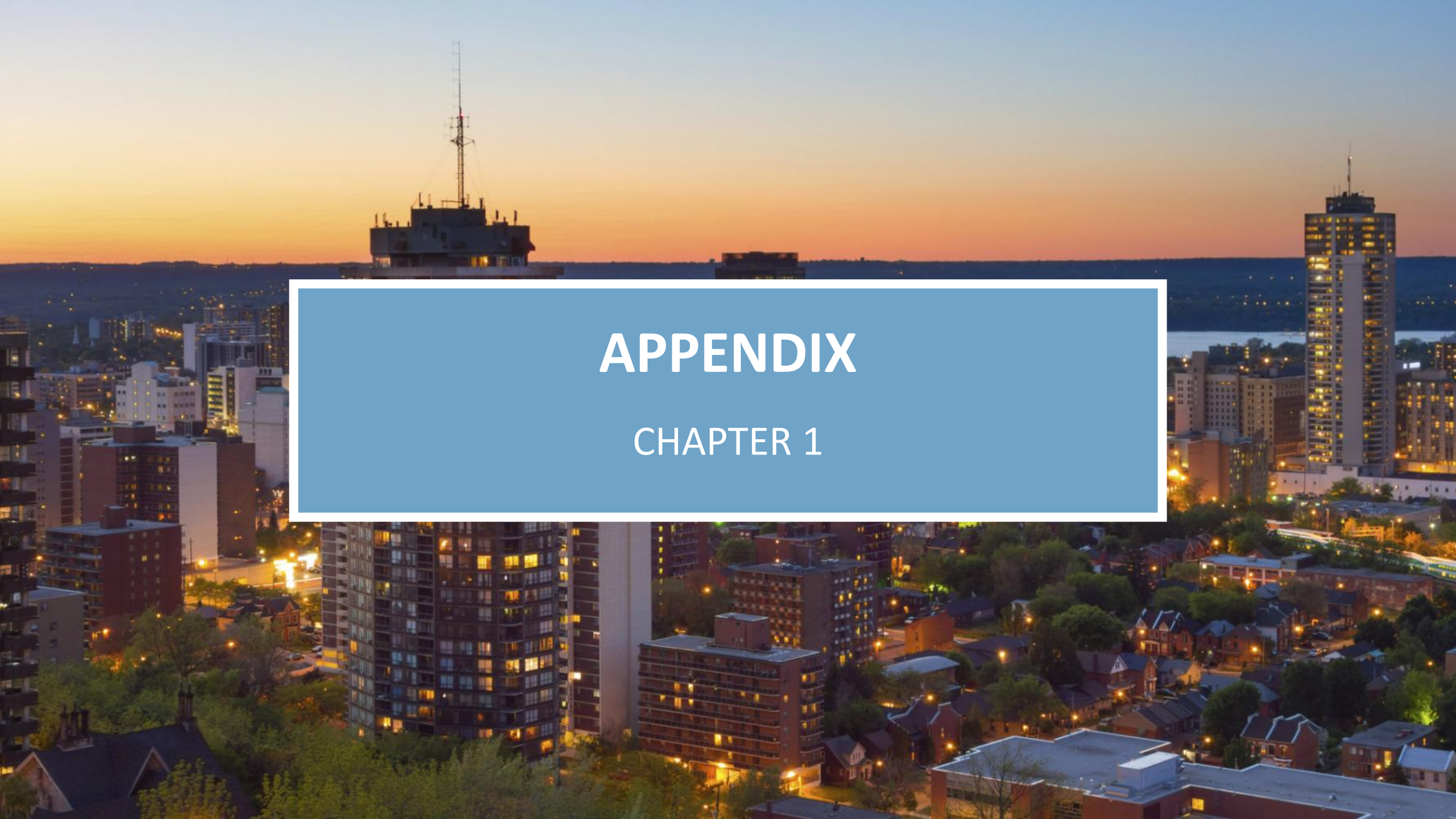
1. Predictive models should remain parsimonious

2. Any uncertainties will be exaggerated in a model

3. Best approach is not always the most feasible

THANKS FOR LISTENING

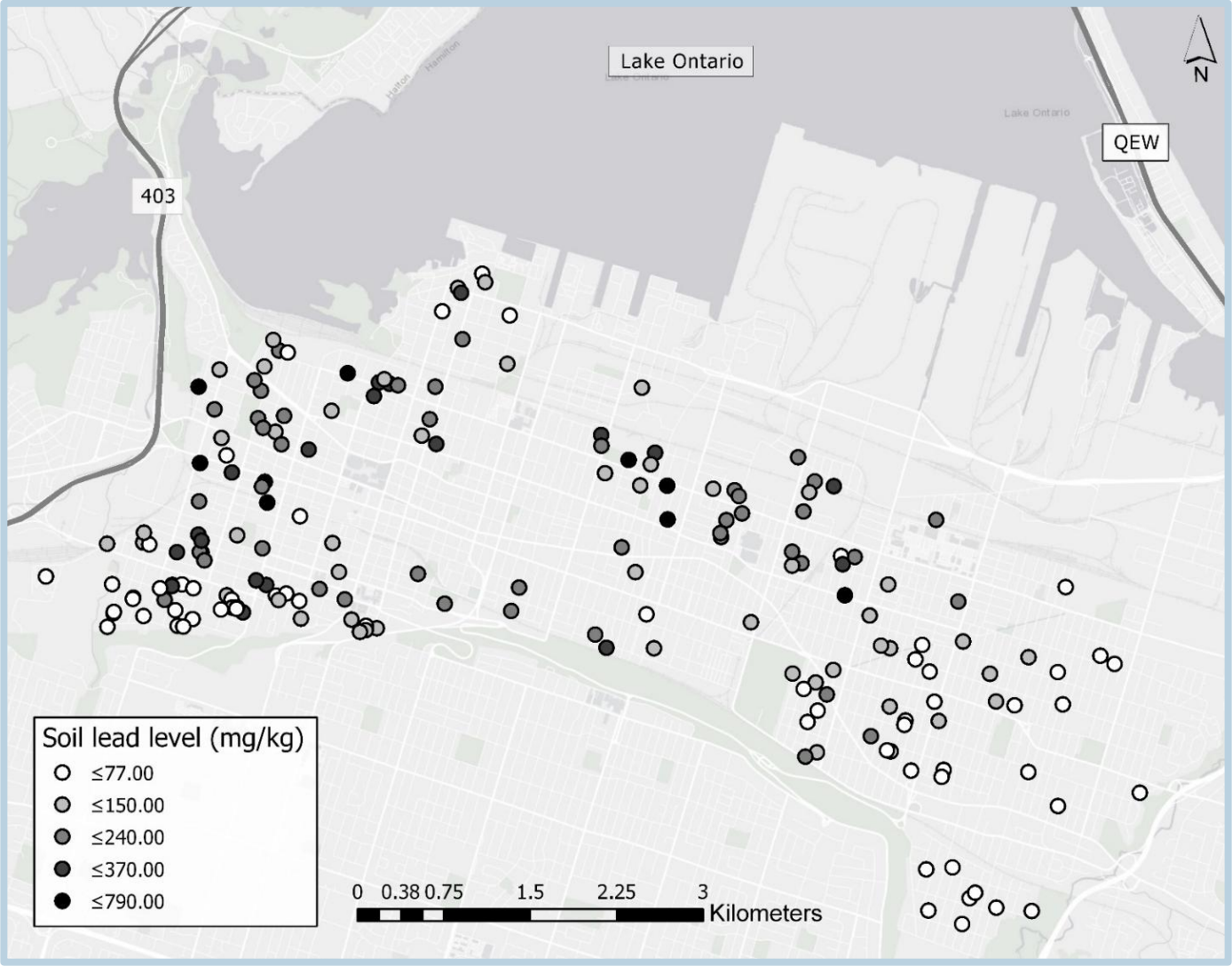
KEVIN MACKAY



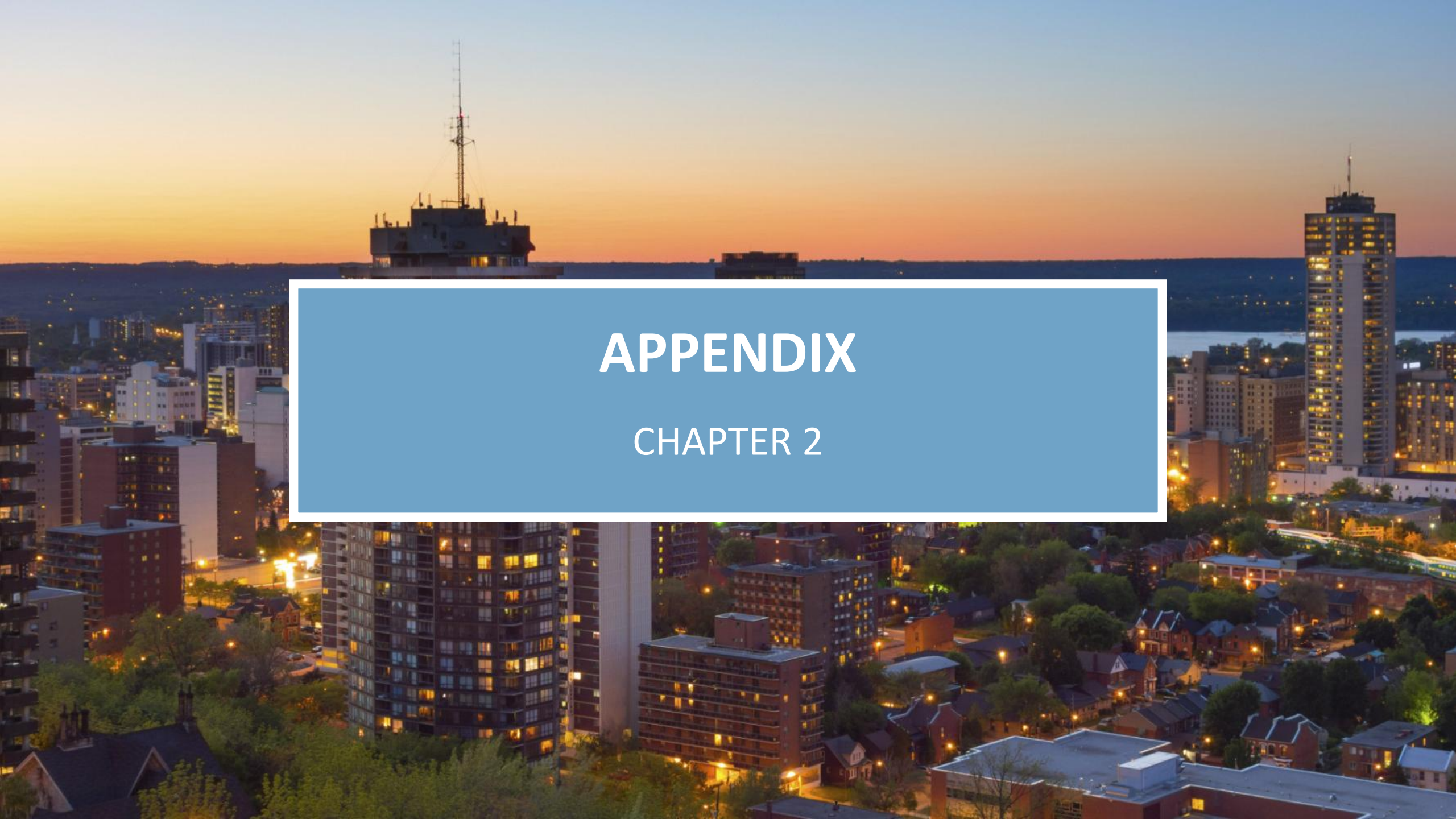
APPENDIX

CHAPTER 1

STUDY AREA


Lake Ontario
QEW
403
N
Soil lead level (mg/kg)
≤77.00
≤150.00
≤240.00
≤370.00
≤790.00
0 0.38 0.75 1.5 2.25 3
Kilometers

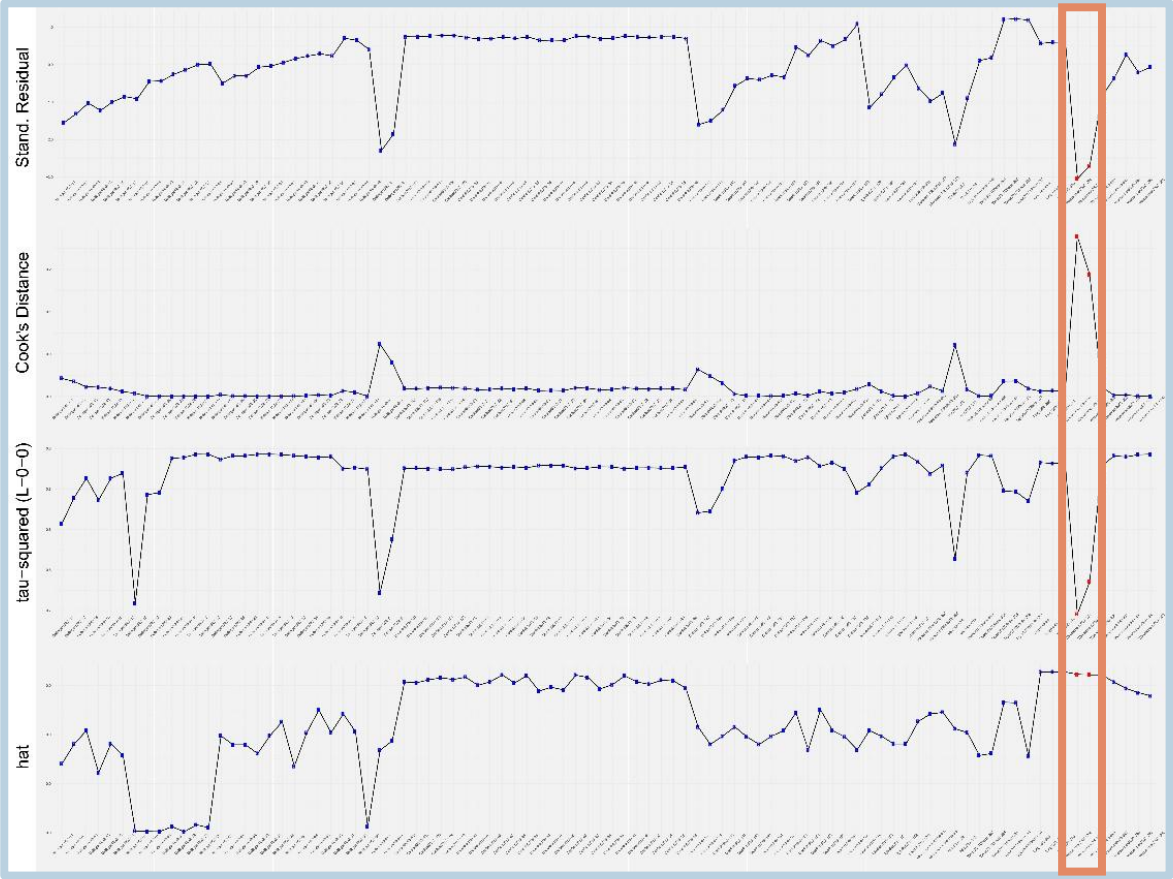
DISTANCE OF HIGHEST CORRELATION WITH SLLs				
INDEPENDENT VARIABLES	YEARS OF DATA	DISTANCE OF HIGHEST CORRELATION [METERS]	SPEARMAN CORRELATION VALUE	*P*-VALUE
LEAD MANUFACTURING	1925-1990	2,010	0.601	<0.001
AUTOMOBILE PAINTERS	1925-1990	1,380	0.587	<0.001
AUTOMOBILE GARAGES	1925-1990	2,260	0.548	<0.001
WHOLESALE TRADE BLDGs	2010	1,360	0.541	<0.001
BATTERY AND SERVICE STATIONS	1925-1990	2,270	0.535	<0.001
JUNK AND SALVAGE YARDS	1925-1990	4,380	0.532	<0.001
PAINT MANUFACTURING	1925-1990	1,990	0.514	<0.001
DENTAL OFFICES	1925-1990	1,700	0.466	<0.001
GASOLINE STATIONS	1925-1990	1,780	0.465	<0.001
RETAIL TRADE	2010	1,440	0.460	<0.001
FINANCE, INSURANCE AND REAL ESTATE	2010	590	0.457	<0.001
PUBLIC ADMINISTRATION BLDGs	2010	470	0.341	<0.001
MAJOR ROADS	1990	170	0.286	<0.001
EXPRESSWAYS	1990	1,090	0.267	<0.001
LOCAL ROADS	1990	30	0.182	0.026
HIGHWAYS	1990	3,220	-0.231	0.005

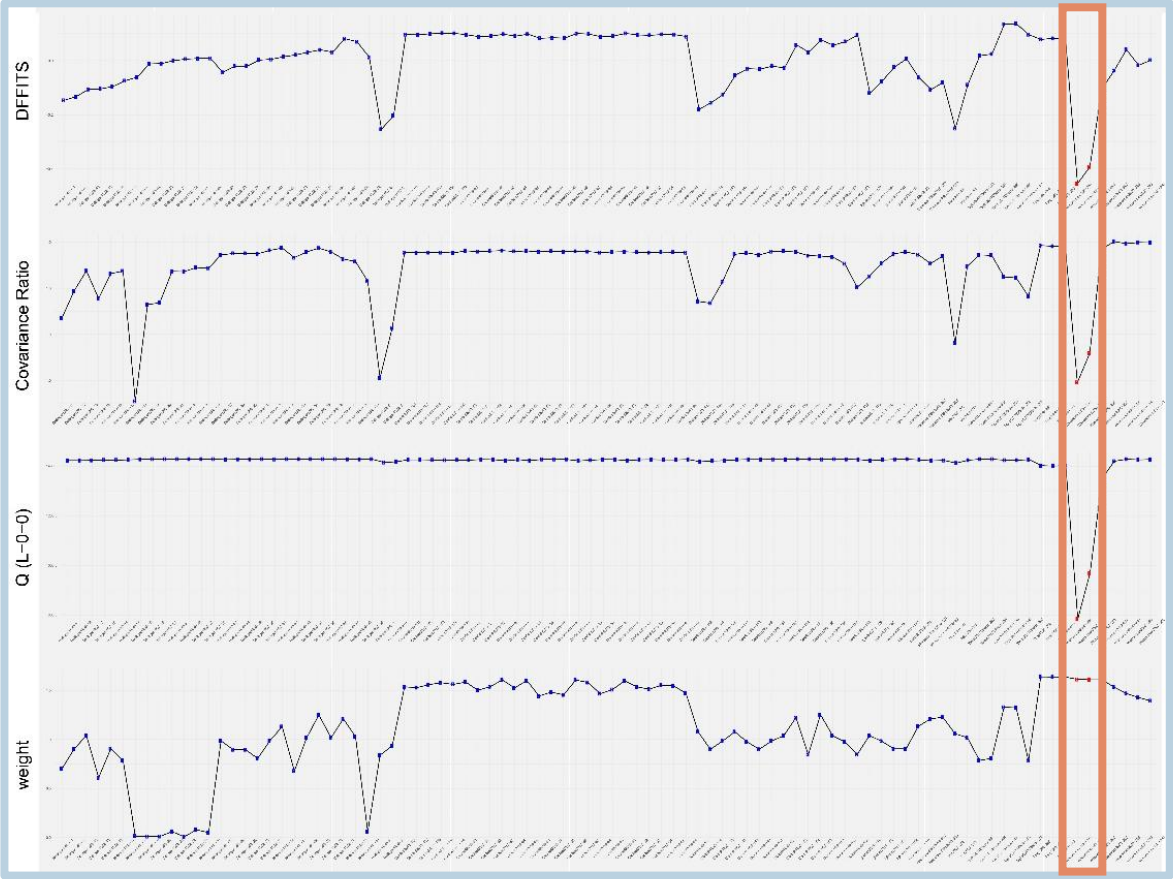


APPENDIX

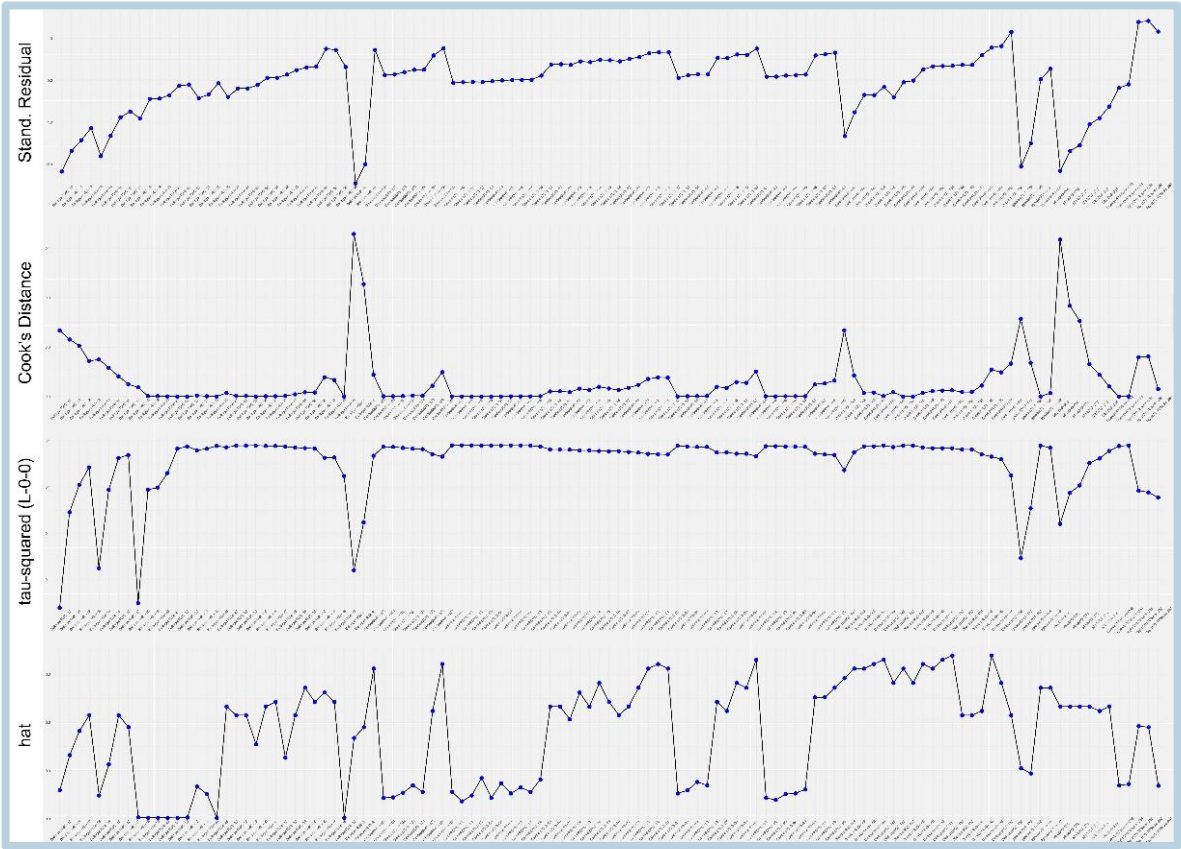
CHAPTER 2

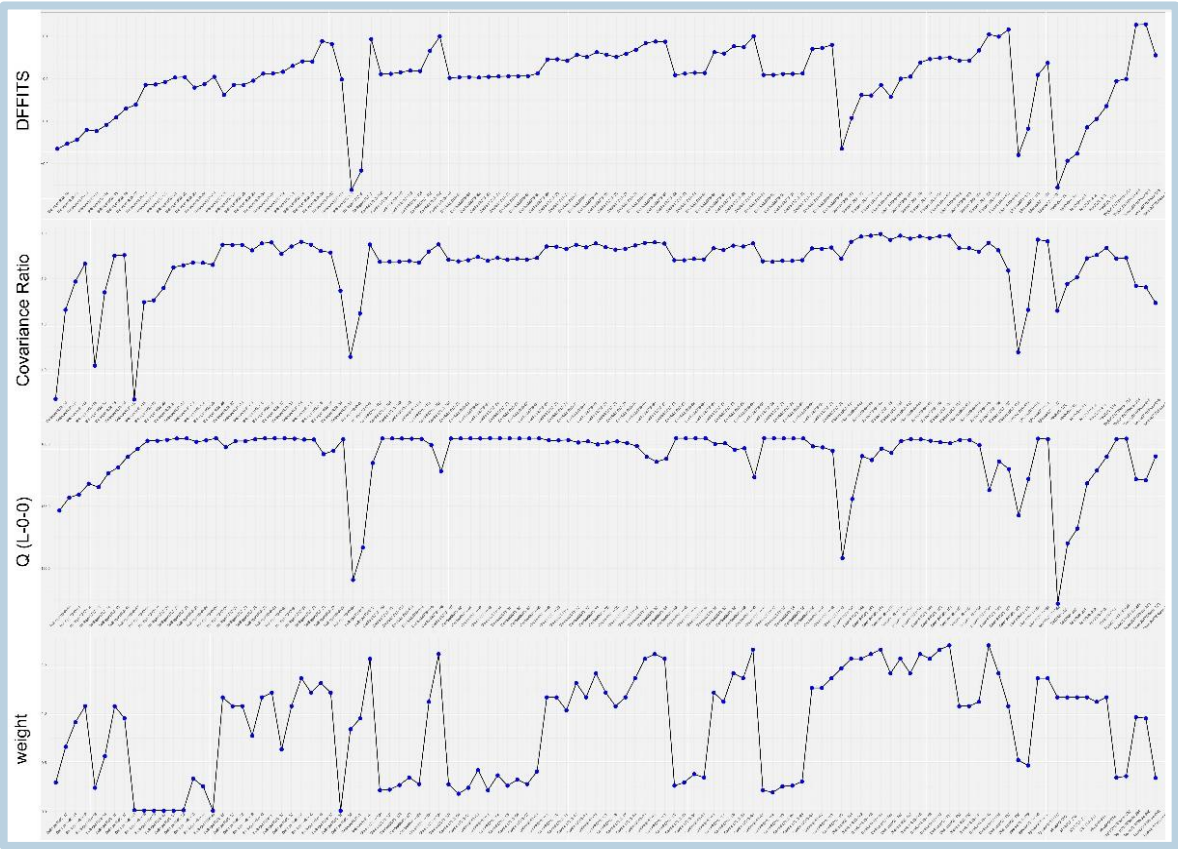
INFLUENCE CHARACTERISTICS NON-LINEAR





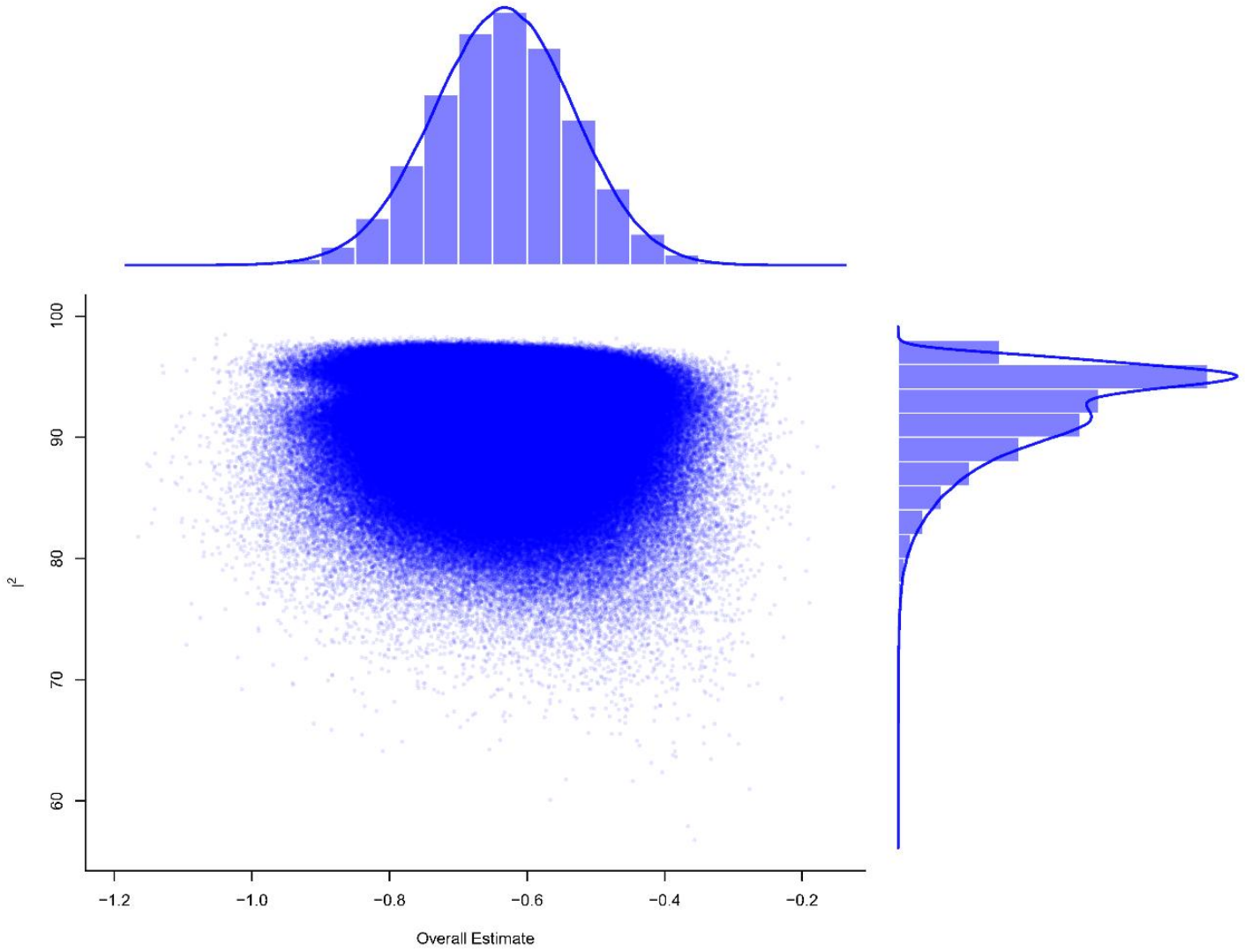
INFLUENCE CHARACTERISTICS LINEAR



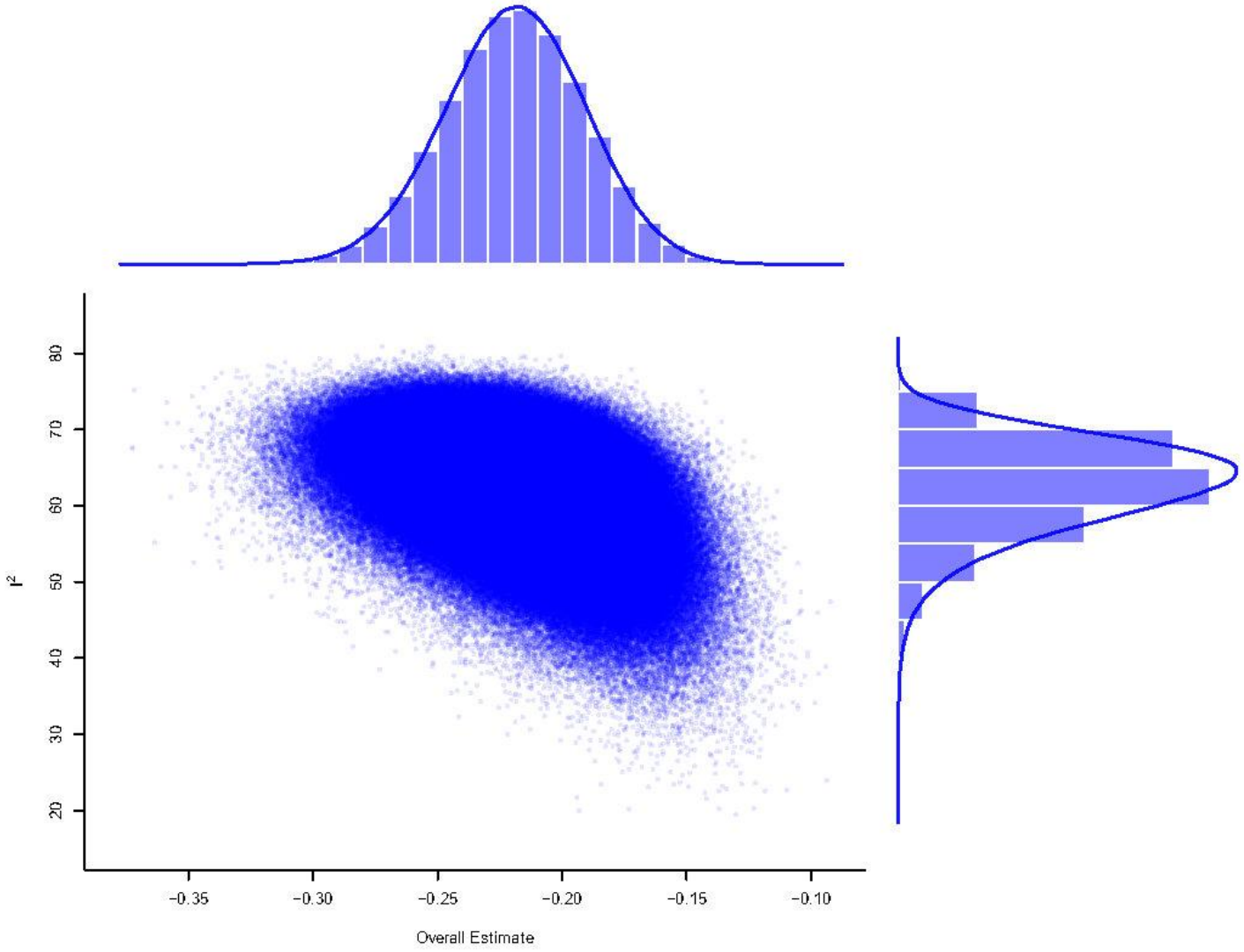


POOLED RESULTS WITH OUTLIERS REMOVED												
		GROUP	POOLED RESULT	95% CONFIDENCE INTERVAL			PREDICTION INTERVAL		HETEROGENEITY MEASURES			
				LOWER	UPPER	*P*	Lower	Upper	τ^2	I^2	Q	*P*
	ALL UBCs	NON-LINEAR PAPERS: 13 UBCs: 139	-0.40	-0.54	-0.27	< 0.0001	-1.96	1.16	0.62	92.4%	1,809	< 0.0001
		LINEAR PAPERS: 7 UBCs: 116	-0.20	-0.26	-0.15	< 0.0001	-0.71	0.30	0.06	47.7%	220	< 0.0001
UBCs REMOVED USING…	CONFIDENCE INTERVALS	NON-LINEAR PAPERS**: 11** **UBCs: 90** **REMOVED: 49**	-0.73	-0.94	-0.53	< 0.0001	-2.53	1.06	0.81	93.8%	1,428	< 0.0001
		LINEAR PAPERS**: 6** **UBCs: 113** **REMOVED: 3**	-0.22	-0.27	-0.17	< 0.0001	-0.70	0.27	0.06	30.2%	160	< 0.01
	INFLUENCE ANALYSIS	NON-LINEAR PAPERS**: 11** **UBCs: 88** **REMOVED: 2**	-0.64	-0.82	-0.45	< 0.0001	-2.29	1.02	0.68	70.7%	297	< 0.0001
		LINEAR PAPERS**: 6** **UBCs: 113** **REMOVED: 0**	-0.22	-0.27	-0.17	< 0.0001	-0.70	0.27	0.06	30.2%	160	< 0.01

GOSH PLOTS
NON-LINEAR



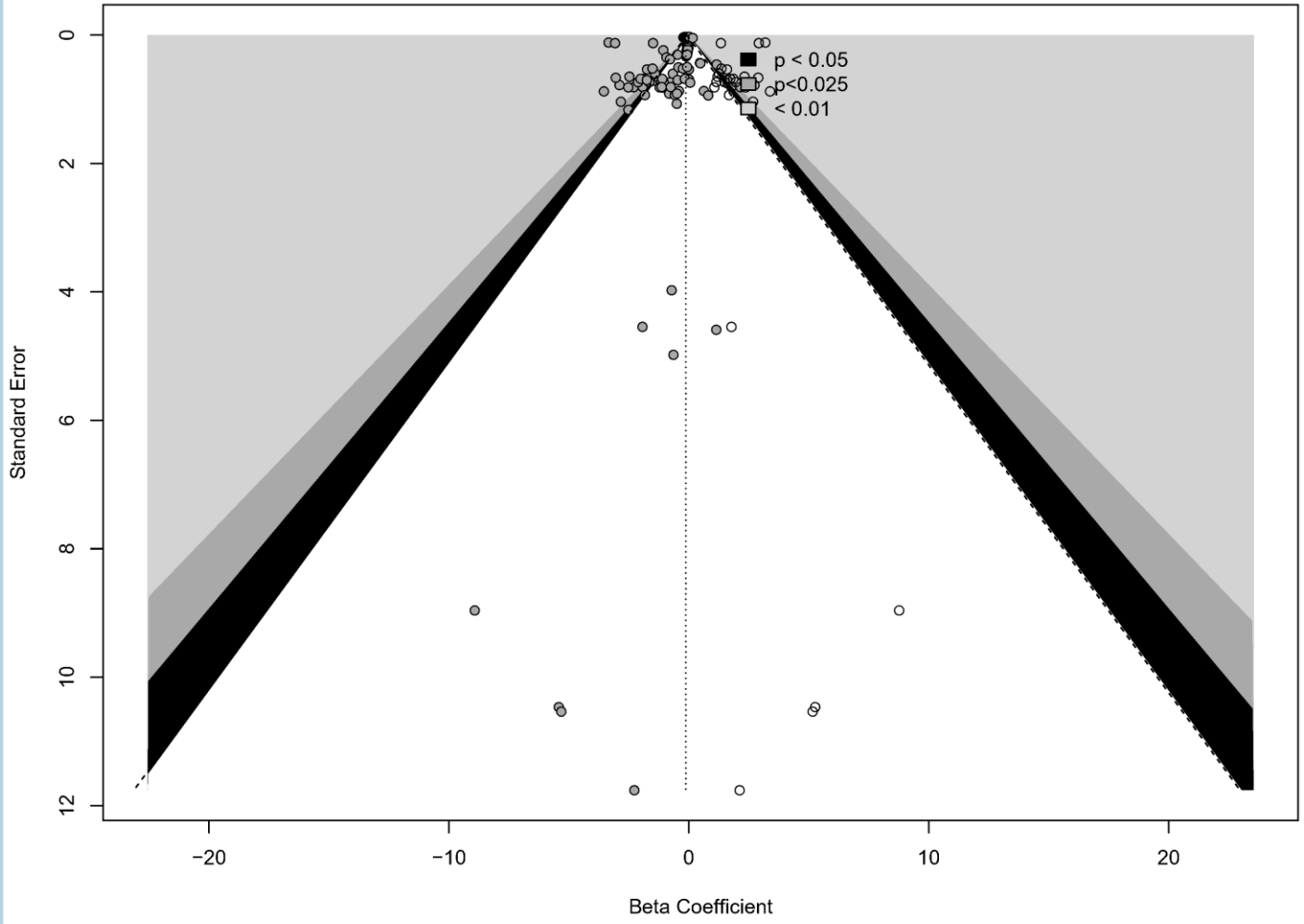
GOSH PLOTS
LINEAR



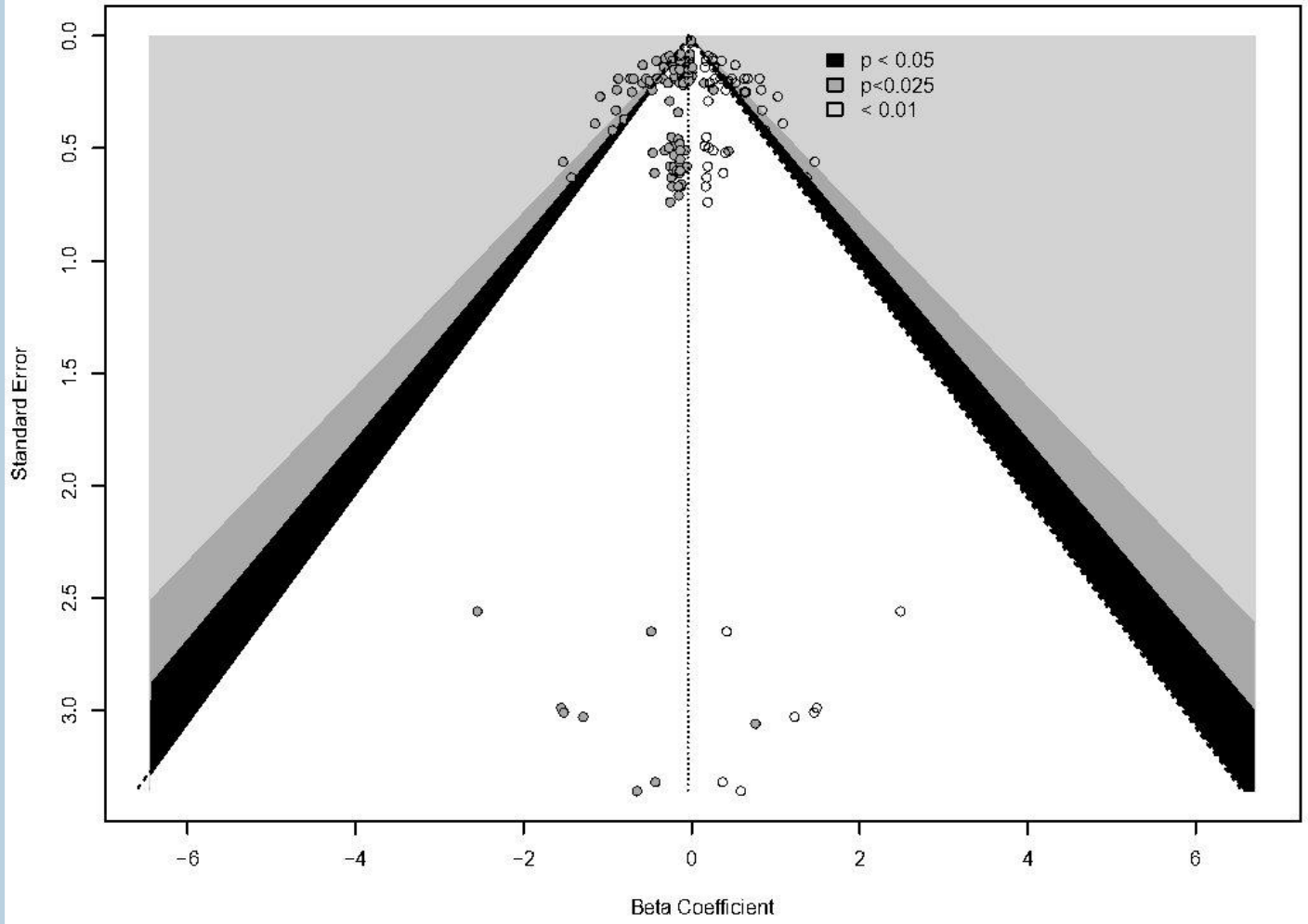
SUBGROUP ANALYSIS									
GROUP	SUBGROUP	POOLED RESULT	95% CONFIDENCE INTERVAL		HETEROGENEITY MEASURES			SUBGROUP DIFFERENCES	
			LOWER	UPPER	τ^2	I^2	Q	Q	*P*
NON-LINEAR	UNADJUSTED PAPERS: 6 UBCs: 39	-0.80	-1.13	-0.47	0.81	63.6%	104	2.16	0.14
	ADJUSTED PAPERS: 10 UBCs: 49	-0.52	-0.73	-0.30	0.53	73.3%	180		
LINEAR	UNADJUSTED PAPERS: 5 UBCs: 54	-0.31	-0.39	-0.22	0.07	41.8%	91	11.10	< 0.01
	ADJUSTED PAPERS: 6 UBCs: 59	-0.14	-0.20	-0.08	0.04	0.0%	54		
NON-LINEAR	INFANT PAPERS: 4 UBCs: 23	-0.22	-0.49	0.04	0.55	0.0%	22	9.50	< 0.01
	TODDLER PAPERS: 4 UBCs: 32	-0.59	-0.90	-0.27	0.63	57.5%	73		
	CHILD PAPERS: 10 UBCs: 33	-0.83	-1.15	-0.52	0.63	82.7%	185		
LINEAR	INFANT PAPERS: 3 UBCs: 21	-0.03	-0.13	0.08	0.07	0.0%	8	15.91	< 0.01
	TODDLER PAPERS: 3 UBCs: 44	-0.19	-0.26	-0.12	0.03	12.3%	49		
	CHILD PAPERS: 5 UBCs: 48	-0.29	-0.38	-0.20	0.07	50.2%	94		

SUBGROUP ANALYSIS									
GROUP	SUBGROUP	POOLED RESULT	95% CONFIDENCE INTERVAL		HETEROGENEITY MEASURES			SUBGROUP DIFFERENCES	
			LOWER	UPPER	τ^2	I^2	Q	Q	*P*
NON-LINEAR	2009-2019 PAPERS: 6 UBCs: 38	-0.29	-0.51	-0.07	0.39	45.0%	67.33	17.48	< 0.0001
	PRE-2009 PAPERS: 5 UBCs: 50	-1.00	-1.26	-0.74	0.70	77.1%	213.86		
LINEAR	2009-2019 PAPERS: 4 UBCs: 63	-0.15	-0.21	-0.09	0.03	12.0%	70	7.31	< 0.01
	PRE-2009 PAPERS: 2 UBCs: 50	-0.29	-0.38	-0.21	0.08	39.1%	81		
NON-LINEAR	G7 PAPERS: 6 UBCs: 74	-0.58	-0.79	-0.36	0.76	50.9%	148.68	2.11	0.15
	OTHERS PAPERS: 5 UBCs: 14	-0.86	-1.21	-0.51	0.25	90.9%	142.40		
LINEAR	G7 PAPERS: 6 UBCs: 109	-0.21	-0.26	-0.16	0.05	28.4%	150.87	1.34	0.25
	OTHERS PAPERS: 1 UBCs: 4	-0.50	-1.28	0.29	0.18	64.8%	8.53		

TRIM-AND-FILL PLOTS
NON-LINEAR



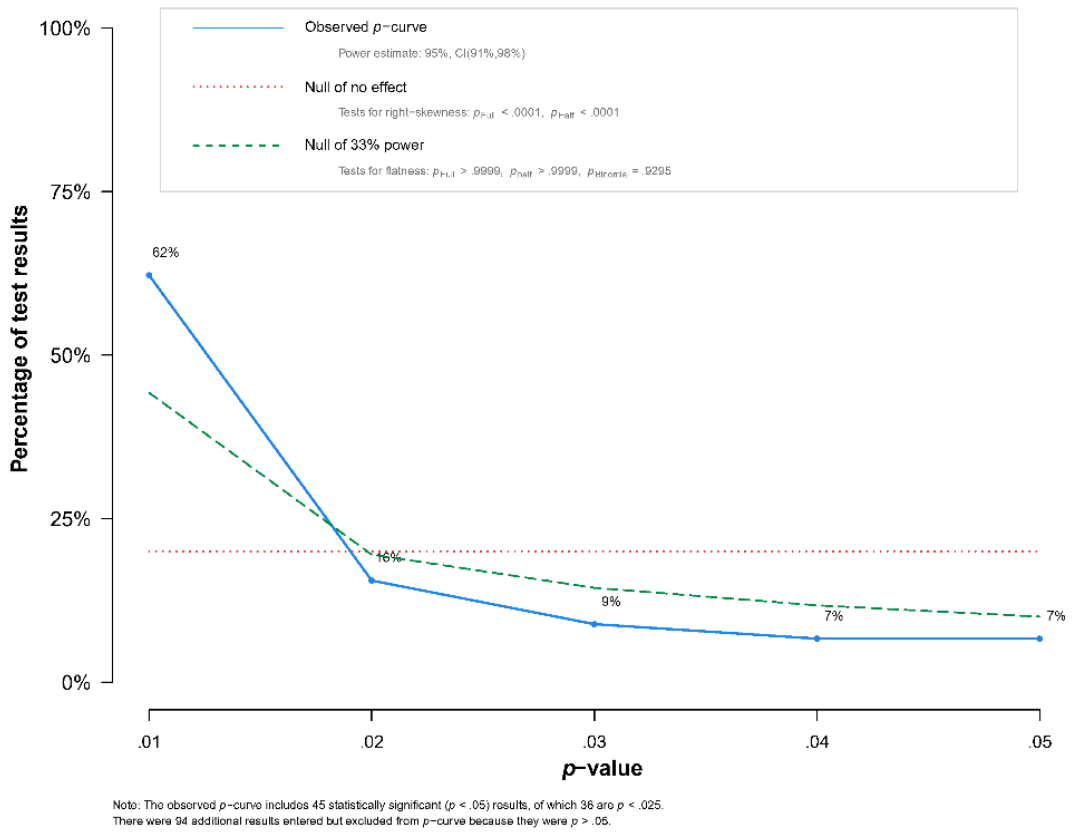
TRIM-AND-FILL PLOTS
LINEAR

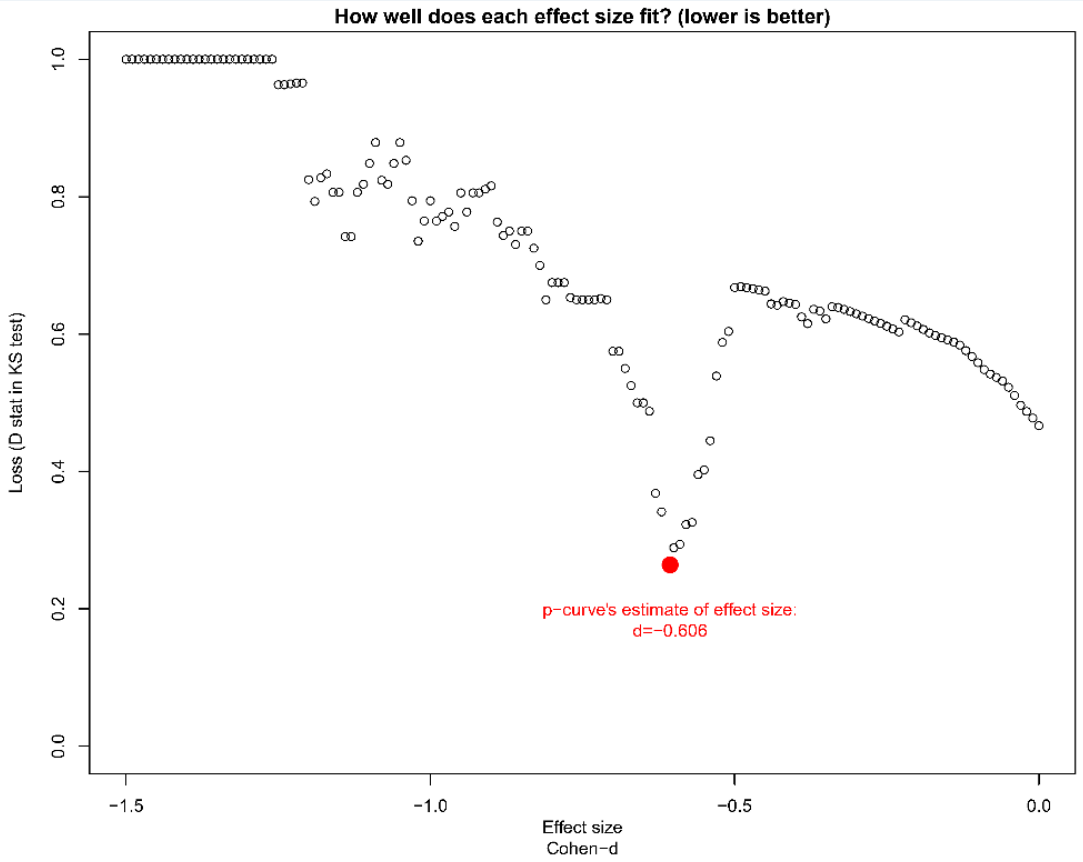


EGGER'S TEST OF SYMMETRY					
GROUP	INTERCEPT	90% CONFIDENCE INTERVAL		T	*P*
		LOWER	UPPER		
NON-LINEAR	-1.33	-2.11	-0.54	-3.29	< 0.01
LINEAR	-0.87	-1.27	-0.48	-5.67	0

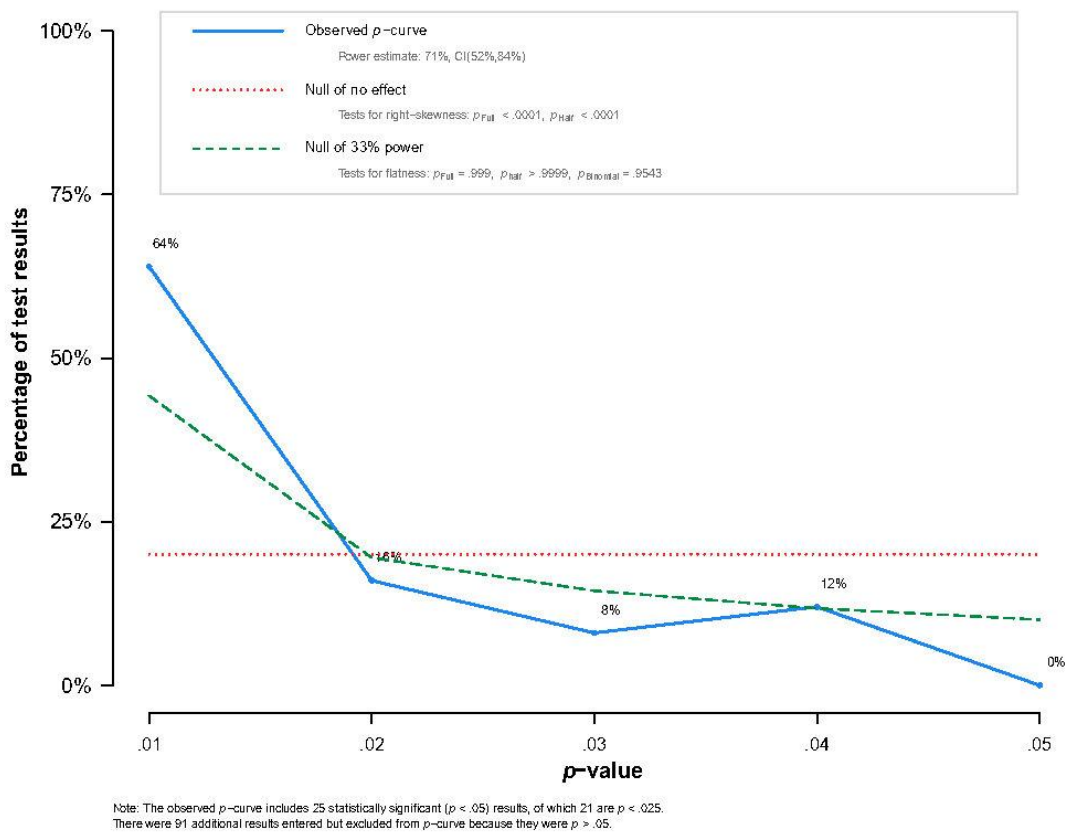
DUVAL AND TWEEDIE'S TRIM AND FILL PROCEDURE										
GROUP	POOLED RESULT	95% CONFIDENCE INTERVAL			PREDICTION INTERVAL		HETEROGENEITY MEASURES			
		LOWER	UPPER	*P*	LOWER	UPPER	τ^2	I^2	Q	*P*
NON-LINEAR PAPERS: 13 UBCs: 168 ADDED: 29	-0.13	-0.31	0.05	0.16	-2.41	2.15	1.32	95.1%	3,433	0
LINEAR PAPERS: 7 UBCs: 163 ADDED: 47	-0.04	-0.11	0.02	0.20	-0.80	0.71	0.14	65.1%	465	< 0.0001

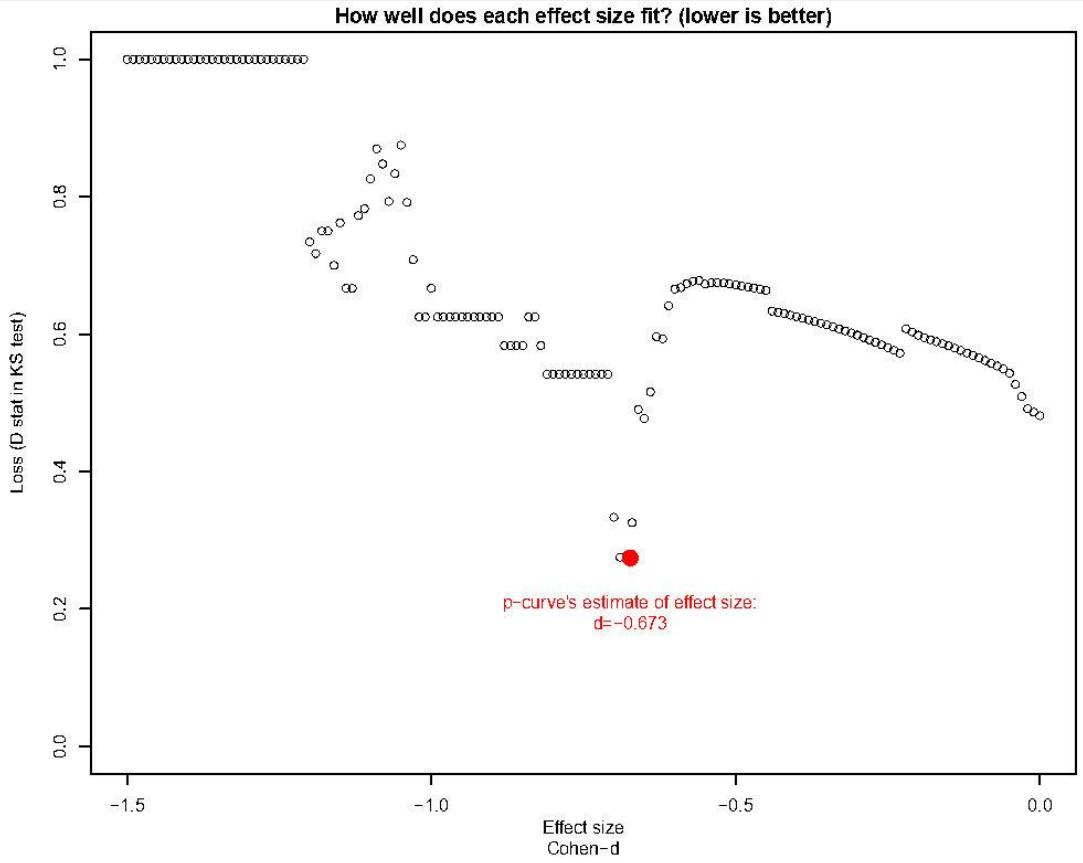
P-CURVE ANALYSIS NON-LINEAR





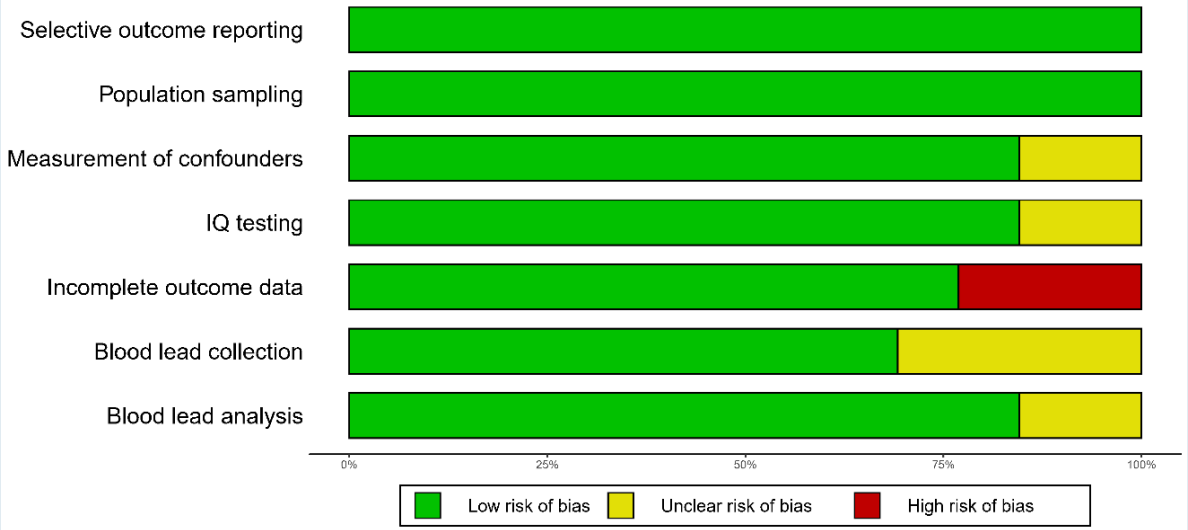
P-CURVE ANALYSIS
LINEAR

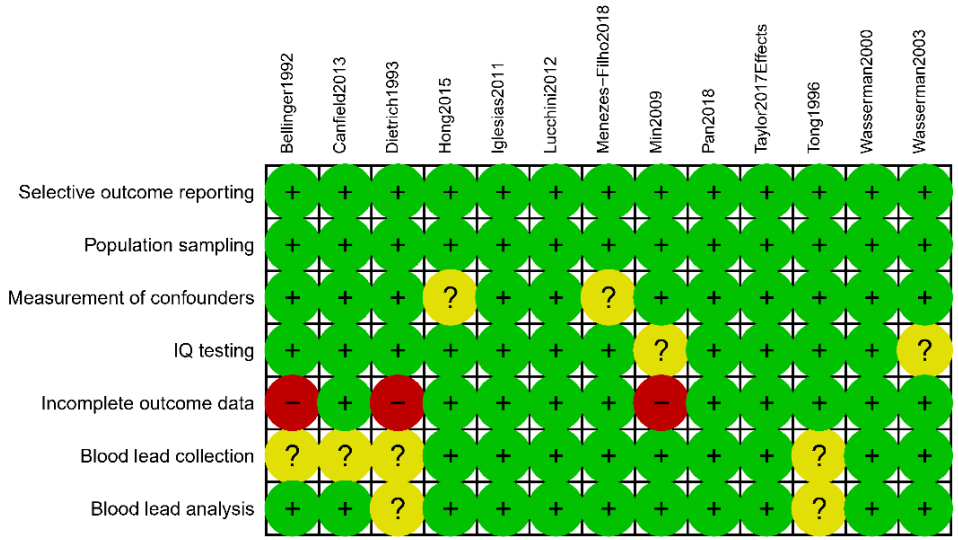


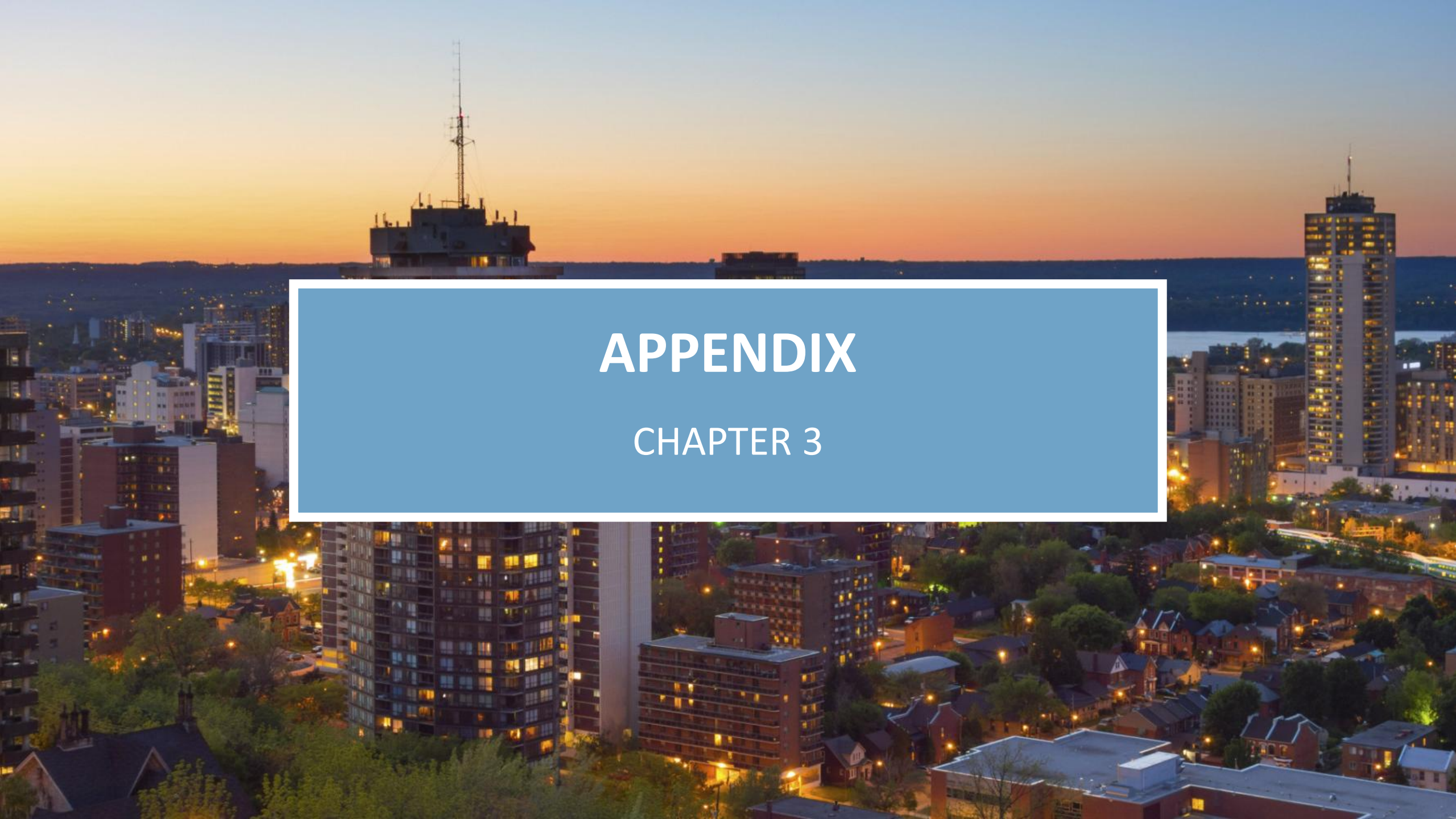


P-CURVE ANALYSIS								
GROUP	TOTAL NUMBER OF UBCS INCLUDED IN THE ANALYSIS		POWER ESTIMATE			EVIDENTIAL VALUE		*P*-CURVE ESTIMATE
			POWER ESTIMATE	95% C.I.		EVIDENTIAL VALUE PRESENT	EVIDENTIAL VALUE ABSENT AND/OR INADEQUATE	
	P<0.05	P<0.025		LOWER	UPPER			
NON-LINEAR ***P* < 0.05: 45** ***P* > 0.05: 94**	45 (34.4%)	36 (25.9%)	95%	91.1%	97.6%	Yes	No	-0.61
LINEAR ***P* < 0.05: 25** ***P* > 0.05: 91**	25 (21.55%)	21 (18.1%)	71%	52%	84.3%	Yes	No	-0.67

RISK OF BIAS SUMMARY







APPENDIX

CHAPTER 3

COST OF LEAD ABATEMENT				
LAND USE	TYPE OF ABATEMENT		TOTAL COST	COST PER CHILD
RESIDENTIAL	SOIL [mg/kg]	> 70	$174,708,760	$31,707
		> 140	$112,322,261	$25,791
	PAINT	SCREENING	$30,620,666	$4,389
		REMOVAL	$275,585,988	$39,510
	LEAD PIPES	LOWER	$47,261,661	$8,100
		UPPER	$70,892,492	$12,149
PARKS [HIGH VOLUME]	SOIL [mg/kg]	> 70	$38,072,010	$1,345
		> 140	$24,165,812	$853
PARKS [LOW VOLUME]	SOIL [mg/kg]	> 70	$18,395,300	$10,663
		> 140	$12,512,398	$9,067
ELEMENTARY SCHOOLS	SOIL [mg/kg]	> 70	$9,875,431	$4,655
		> 140	$7,626,521	$4,566
	PAINT	SCREENING	$28,333	$16
		REMOVAL	$255,000	$147
	LEAD PIPES	LOWER	$14,319	$52
		UPPER	$21,478	$78
NURSERY- AND PRE-SCHOOLS	SOIL [mg/kg]	> 70	$43,941	$8
		> 140	$43,941	$8

DISCOUNTED COST VS. RETURN ON INVESTMENT

LAND USE	TYPE OF ABATEMENT		TOTAL COST WITH DISCOUNT RATE		RETURN ON INVESTMENT [PER DOLLAR SPENT]	
			3%	8%	CONSERVATIVE	OPTIMISTIC
RESIDENTIAL	SOIL [MG/KG]	> 70	$812,550,508	$9,557,624,559	$712,811,742	$9,266,552,652
		> 140	$522,398,019	$6,144,706,197	$458,274,822	$5,957,572,694
	PAINT	SCREENING	$142,413,224	$1,675,135,405	$623,855,488	$8,110,121,336
		REMOVAL	$1,281,718,985	$15,076,218,313	$5,614,699,517	$72,991,093,721
	LEAD PIPES	LOWER	$219,808,593	$2,585,498,357	$48,206,894	$626,689,622
		UPPER	$329,712,891	$3,878,247,562	$72,310,340	$940,034,433
PARKS [HIGH VOLUME]	SOIL [MG/KG]	> 70	$177,068,574	$2,082,768,934	$155,333,799	$2,019,339,393
		> 140	$112,392,434	$1,322,015,898	$98,596,513	$1,281,754,662
PARKS [LOW VOLUME]	SOIL [MG/KG]	> 70	$85,554,441	$1,006,334,033	$75,052,827	$975,686,743
		> 140	$58,193,735	$684,503,756	$51,050,584	$663,657,592
ELEMENTARY SCHOOLS	SOIL [MG/KG]	> 70	$45,929,503	$540,245,732	$40,291,758	$523,792,861
		> 140	$35,470,079	$417,216,769	$31,116,204	$404,510,645
	PAINT	SCREENING	$131,774	$1,549,986	$337,167	$4,383,175
		REMOVAL	$1,185,976	$13,950,040	$3,034,506	$39,448,574
	LEAD PIPES	LOWER	$66,596	$783,336	$13,611	$189,867
		UPPER	$99,892	$1,174,976	$21,907	$284,800
NURSERY- AND PRE-SCHOOLS	SOIL [MG/KG]	> 70	$204,365	$2,403,838	$179,277	$2,330,609
		> 140	$204,365	$2,403,838	$179,277	$2,330,609

HAZARD QUOTIENT				
AGE GROUP	52 WEEKS OF EXPOSURE		40 WEEKS OF EXPOSURE	
	MAXIMUM	MEAN	MAXIMUM	MEAN
INFANT 0 - 0.5 years	0.396	0.195	0.304	0.150
TODDLER 0.6 - 4 years	0.786	0.388	0.605	0.299
CHILD 5 - 11 years	0.099	0.049	0.076	0.037
HQ < 0.5	VERY UNLIKELY TO EXCEED 5 µg/dL			
0.5 > HQ > 1.0	UNLIKELY TO EXCEED 5 µg/dL			
HQ > 1.0	MAY EXCEED 5 µg/dL			

HAZARD QUOTIENT CALCULATIONS

FORMULAS

$$\textbf{Hazard Quotient} = \frac{\text{Estimated Exposure [Dose] [mg/kg bw/day]}}{\text{Tolerable Daily Intake [TDI] [mg/kg bw/day]}}$$

$$\textbf{Estimated Exposure [Dose] = [mg/kg bw/day]} \quad \frac{C_s * IR_s * RAF_{oral} * D_2 * D_3}{BW}$$

Where:
C_s = Concentration in soil (mg/kg)
IR_s = Soil ingestion rate (kg/day)
RAF_{oral} = Relative absorption factor from the gastrointestinal tract
D_2 = Days per week exposed (divided by 7 days)
D_3 = Weeks per year exposed (divided by 52 weeks)
BW = Body weight (kg)

TDI relating to a blood lead level of 5 µg/dL = 0.0006 * 2.5 = 0.0015 mg/kg bw/day

ASSUMPTIONS

CHARACTERISTIC	INFANT	TODDLER	CHILD
AGE	0 – 6 mo.	7 mo. – 4 yr.	5 – 11 yr.
C_s [mg/kg]	SLL varies across space	SLL varies across space	SLL varies across space
IR_s [kg/day]	0.00002	0.00008	0.00002
RAF_{oral}	1.0	1.0	1.0
BW [kg]	8.2	16.5	32.9

EXAMPLE

GIVEN:
C_s = 250 mg/kg [Soil lead level]
IR_s = 0.00008 kg/day [Soil ingestion rate]
RAF_{oral} = 1 [Relative absorption factor]
D_2 = 7 days / 7 days = 1 [Days per week exposed]
D_3 = 40 weeks / 52 weeks = 0.769 [Weeks per year exposed]
BW = 16.5 kg [Body weight]

TDI relating to a blood lead level of 5 µg/dL = 0.0015 mg/kg bw/day

$$\textbf{Estimated Exposure [Dose] = [mg/kg bw/day]} \quad \frac{\textbf{250 * 0.00008 * 1 * 1 * [40/52]}}{16.5} \quad \textbf{= 0.00093240093}$$

$$\text{Hazard Quotient} = \frac{0.00093240093}{0.0015} \quad = 0.62160062$$

Therefore, a toddler is unlikely to exceed the threshold of 5 µg/dL and suffer adverse health effects